

1001 PHƯƠNG TRÌNH PHẢN ỨNG HÓA HỌC PHỨC TẠP NHẤT

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1. $2 \text{KMnO}_4 + 10 \text{FeSO}_4 + 8 \text{H}_2\text{SO}_4 \rightarrow 2 \text{MnSO}_4 + 5 \text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 8 \text{H}_2\text{O}$
2. $2 \text{KMnO}_4 + 4 \text{K}_2\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{MnO}_2 + \text{K}_2\text{SO}_4 + \text{KOH}$
3. $2 \text{KMnO}_4 + \text{K}_2\text{SO}_3 + 2 \text{KOH} \rightarrow 2 \text{K}_2\text{MnO}_4 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
4. $\text{K}_2\text{Cr}_2\text{O}_7 + 6 \text{FeSO}_4 + 7 \text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + 3 \text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 7 \text{H}_2\text{O}$
5. $2 \text{KCrO}_4 + 3(\text{NH}_4)_2\text{S} + 2 \text{H}_2\text{O} \rightarrow 2 \text{Cr}(\text{OH})_3 + 3 \text{S} + 6 \text{NH}_3 + 4 \text{KOH}$
6. $\text{Fe} + 6 \text{HNO}_3 (\text{đ. nóng}) \rightarrow \text{Fe}(\text{NO}_3)_3 + 3 \text{NO}_2 + 3 \text{H}_2\text{O}$
7. $\text{Fe} + 4 \text{HNO}_3 (\text{l}) \rightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO} + 2 \text{H}_2\text{O}$
8. $3 \text{Cu} + 2 \text{NO}_3^- + 8 \text{H}^+ \rightarrow 3 \text{Cu}^{2+} + 2 \text{NO} + 4 \text{H}_2\text{O}$
9. $\text{Cu} + 2 \text{H}_2\text{SO}_4 (\text{đ. nóng}) \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2 \text{H}_2\text{O}$
10. $2 \text{Fe} + 6 \text{H}_2\text{SO}_4 (\text{đ. nóng}) \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3 \text{SO}_2 + 6 \text{H}_2\text{O}$
11. $\text{Fe} + 4 \text{HNO}_3 (\text{l}) \rightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO} + 2 \text{H}_2\text{O}$
12. $\text{Fe} + 6 \text{HNO}_3 (\text{đ. nóng}) \rightarrow \text{Fe}(\text{NO}_3)_3 + 3 \text{NO}_2 + 3 \text{H}_2\text{O}$
13. $3 \text{Cu} + 2 \text{NO}_3^- + 8 \text{H}^+ \longrightarrow 3 \text{Cu}^{2+} + 2 \text{NO} + 4 \text{H}_2\text{O}$
14. $2 \text{Fe} + 6 \text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3 \text{SO}_2 + 6 \text{H}_2\text{O}$
15. $\text{CrSO}_4 + \text{O}_2 + \text{H}_2\text{SO}_4 \rightarrow 2 \text{Cr}_2(\text{SO}_4)_3 + 2 \text{H}_2\text{O}$
16. $2 \text{Cr} + 6 \text{H}_2\text{SO}_4 \rightarrow 2 \text{Cr}_2(\text{SO}_4)_3 + 3 \text{SO}_2 \uparrow + 3 \text{H}_2\text{O}$
17. $\text{Cr} + 4 \text{HNO}_3 \rightarrow \text{Cr}(\text{NO}_3)_3 + \text{NO} \uparrow + 2 \text{H}_2\text{O}$
18. $\text{Cr} + \text{HNO}_3 + 3 \text{HCl} \rightarrow \text{CrCl}_3 + \text{NO} \uparrow + 2 \text{H}_2\text{O}$
19. $4 \text{FeCr}_2\text{O}_4 + 8 \text{Na}_2\text{CO}_3 + 7 \text{O}_2 \rightarrow 8 \text{Na}_2\text{CrO}_4 + 2 \text{Fe}_2\text{O}_3 + 8 \text{CO}_2$
20. $2 \text{Na}_2\text{CrO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
21. $\text{Na}_2\text{Cr}_2\text{O}_7 + 2 \text{C} \rightarrow \text{Cr}_2\text{O}_3 + \text{Na}_2\text{CO}_3 + \text{CO}$
22. $\text{Cr}_2\text{O}_3 + 2 \text{Al} \xrightarrow{t^\circ} 2 \text{Cr} + \text{Al}_2\text{O}_3$
23. $4 \text{CrCl}_2 + \text{O}_2 + 4 \text{HCl} \rightarrow 4 \text{CrCl}_3 + 2 \text{H}_2\text{O}$
24. $4 \text{Cr}(\text{OH})_2 + \text{O}_2 + 2 \text{H}_2\text{O} \rightarrow 4 \text{Cr}(\text{OH})_3$
25. $4 \text{Cr}(\text{OH})_2 + \text{O}_2 + 2 \text{H}_2\text{O} \longrightarrow 4 \text{Cr}(\text{OH})_3$
26. $\text{Cr}(\text{OH})_2 + 2 \text{HCl} \longrightarrow \text{CrCl}_2 + 2 \text{H}_2\text{O}$
27. $2 \text{CrCl}_2 + \text{Cl}_2 \longrightarrow 2 \text{CrCl}_3$
28. $\text{Cr}_2\text{O}_3 + 6 \text{HCl} \longrightarrow 2 \text{CrCl}_3 + 3 \text{H}_2\text{O}$
29. $\text{Cr}_2\text{O}_3 + 2 \text{NaOH} \longrightarrow 2 \text{NaCrO}_2 + \text{H}_2\text{O}$
30. $\text{Cr}_2\text{O}_3 + 2 \text{NaOH} + 3 \text{H}_2\text{O} \longrightarrow 2 \text{Na}[\text{Cr}(\text{OH})_4]$
31. $\text{Cr}(\text{OH})_3 + 3 \text{HCl} \longrightarrow \text{CrCl}_3 + 3 \text{H}_2\text{O}$
32. $\text{Cr}(\text{OH})_3 + \text{NaOH} \longrightarrow \text{Na}[\text{Cr}(\text{OH})_4]$
33. $\text{Cr}(\text{OH})_3 + \text{NaOH} \longrightarrow \text{NaCrO}_2 + 2 \text{H}_2\text{O}$
34. $2 \text{Cr}(\text{OH})_3 \rightarrow \text{Cr}_2\text{O}_3 + 3 \text{H}_2\text{O}$
35. $\text{Cr}(\text{OH})_3 + 3 \text{Na}_2\text{O}_2 \rightarrow 2 \text{Na}_2\text{CrO}_4 + 2 \text{NaOH} + 2 \text{H}_2\text{O}$
36. $2 \text{Cr}(\text{OH})_3 + 3 \text{H}_2\text{O}_2 + 4 \text{NaOH} \rightarrow 2 \text{Na}_2\text{CrO}_4 + 8 \text{H}_2\text{O}$
37. $2 \text{Cr}(\text{OH})_3 + 3 \text{Cl}_2 + 10 \text{NaOH} \rightarrow 2 \text{Na}_2\text{CrO}_4 + 6 \text{NaCl} + 8 \text{H}_2\text{O}$
38. $2 \text{Cr}(\text{OH})_3 + 3 \text{Br}_2 + 10 \text{NaOH} \rightarrow 2 \text{Na}_2\text{CrO}_4 + 6 \text{NaBr} + 8 \text{H}_2\text{O}$
39. $2 \text{Cr}(\text{OH})_3 + 3 \text{NaOCl} + 4 \text{NaOH} \rightarrow 2 \text{Na}_2\text{CrO}_4 + 3 \text{NaCl} + 5 \text{H}_2\text{O}$
40. $2 \text{Cr}(\text{OH})_3 + 3 \text{PbO}_2 + 4 \text{NaOH} \rightarrow 2 \text{Na}_2\text{CrO}_4 + 3 \text{PbO} + 5 \text{H}_2\text{O}$
41. $\text{Cr}(\text{OH})_3 + 3 \text{KMnO}_4 + 5 \text{KOH} \rightarrow \text{K}_2\text{CrO}_4 + 3 \text{K}_2\text{MnO}_4 + 4 \text{H}_2\text{O}$

42. $\text{CrCl}_3 + 3\text{NaOH} \rightarrow \text{Cr}(\text{OH})_3\downarrow + 3\text{NaCl}$
43. $\text{Cr}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaCrO}_2 + 2\text{H}_2\text{O}$
44. $2\text{NaCrO}_2 + 3\text{Na}_2\text{O}_2 + 4\text{H}_2\text{O} \rightarrow 2\text{Na}_2\text{CrO}_4 + 4\text{NaOH}$
45. $2\text{CrCl}_3 + \text{Zn} \longrightarrow 2\text{CrCl}_2 + \text{ZnCl}_2$
46. $\text{Cr}_2(\text{SO}_4)_3 + \text{Zn} \longrightarrow 2\text{CrSO}_4 + \text{ZnSO}_4$
47. $2\text{CrBr}_3 + 3\text{Br}_2 + 16\text{KOH} \rightarrow 2\text{K}_2\text{CrO}_4 + 12\text{KBr} + 8\text{H}_2\text{O}$
48. $2\text{CrCl}_3 + 3\text{Br}_2 + 16\text{KOH} \rightarrow 2\text{K}_2\text{CrO}_4 + 6\text{KBr} + 6\text{KCl} + 8\text{H}_2\text{O}$
49. $\text{Cr}_2(\text{SO}_4)_3 + 3\text{Br}_2 + 16\text{KOH} \rightarrow 2\text{K}_2\text{CrO}_4 + 6\text{KBr} + 3\text{K}_2\text{SO}_4 + 8\text{H}_2\text{O}$
50. $2\text{Cr}(\text{NO}_3)_3 + 3\text{Br}_2 + 16\text{KOH} \rightarrow 2\text{K}_2\text{CrO}_4 + 6\text{KBr} + 6\text{KNO}_3 + 8\text{H}_2\text{O}$
51. $2\text{Cr}^{3+} + 3\text{Br}_2 + 16\text{OH}^- \longrightarrow 2\text{CrO}_4^{2-} + 6\text{Br}^- + 8\text{H}_2\text{O}$
52. $4\text{CrO}_3 + 3\text{S} \longrightarrow 3\text{SO}_2 + 2\text{Cr}_2\text{O}_3$
53. $10\text{CrO}_3 + 6\text{P} \longrightarrow 3\text{P}_2\text{O}_5 + 5\text{Cr}_2\text{O}_3$
54. $4\text{CrO}_3 + 3\text{C} \longrightarrow 3\text{CO}_2 + 2\text{Cr}_2\text{O}_3$
55. $\text{C}_2\text{H}_5\text{OH} + 4\text{CrO}_3 \longrightarrow 2\text{CO}_2 + 3\text{H}_2\text{O} + 2\text{Cr}_2\text{O}_3$
56. $2\text{CrO}_3 + 2\text{NH}_3 \longrightarrow \text{Cr}_2\text{O}_3 + \text{N}_2 + 3\text{H}_2\text{O}$
57. $2\text{K}_2\text{CrO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{Cr}_2\text{O}_7 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
58. $\text{K}_2\text{Cr}_2\text{O}_7 + 2\text{KOH} \longrightarrow 2\text{K}_2\text{CrO}_4 + \text{H}_2\text{O}$
59. $\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{FeSO}_4 + 7\text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + 3\text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$
60. $\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{KI} + 7\text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + 4\text{K}_2\text{SO}_4 + 3\text{I}_2 + 7\text{H}_2\text{O}$
61. $\text{K}_2\text{Cr}_2\text{O}_7 + 14\text{HCl} \rightarrow 2\text{KCl} + 3\text{CrCl}_3 + 3\text{Cl}_2 + 7\text{H}_2\text{O}$
62. $\text{K}_2\text{Cr}_2\text{O}_7 + 3\text{H}_2\text{S} + 4\text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O} + 3\text{S}$
63. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{t^\circ} \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$
64. $\text{Cr}_2(\text{SO}_4)_3 + 6\text{KOH} \rightarrow 2\text{Cr}(\text{OH})_3 + 3\text{K}_2\text{SO}_4$
65. $2\text{Cr}(\text{OH})_3 + 3\text{Br}_2 + 10\text{KOH} \rightarrow 2\text{K}_2\text{CrO}_4 + 6\text{KBr} + 8\text{H}_2\text{O}$
66. $2\text{K}_2\text{CrO}_4 + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{Cr}_2\text{O}_7 + \text{K}_2\text{SO}_4$
67. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \text{ đặc} \rightarrow \text{CrO}_3 + \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$
68. $2\text{Fe} + 6\text{H}_2\text{SO}_4 \text{ (đặc)} \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + 3\text{SO}_2\uparrow + 6\text{H}_2\text{O}$
69. $\text{Fe} + 6\text{HNO}_3 \text{ (đặc)} \xrightarrow{t^\circ} \text{Fe}(\text{NO}_3)_3 + 3\text{NO}_2\uparrow + 3\text{H}_2\text{O}$
70. $\text{Fe} + 4\text{HNO}_3 \text{ (loãng)} \xrightarrow{t^\circ} \text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + 2\text{H}_2\text{O}$
71. $\text{Fe} + \text{CuSO}_4 \longrightarrow \text{FeSO}_4 + \text{Cu}\downarrow$
72. $2\text{FeO} + 4\text{H}_2\text{SO}_4 \text{ (đặc)} \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2\uparrow + 4\text{H}_2\text{O}$
73. $3\text{FeO} + 10\text{HNO}_3 \text{ (loãng)} \xrightarrow{t^\circ} 3\text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + 5\text{H}_2\text{O}$
74. $\text{FeO} + \text{H}_2 \xrightarrow{t^\circ} \text{Fe} + \text{H}_2\text{O}$
75. $\text{Fe}_2\text{O}_3 + \text{CO} \xrightarrow{500-600^\circ\text{C}} 2\text{FeO} + \text{CO}_2$
76. $4\text{Fe}(\text{OH})_2 + \text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 4\text{Fe}(\text{OH})_3$
77. $4\text{Fe}(\text{OH})_2 + \text{O}_2 \xrightarrow{t^\circ} 2\text{Fe}_2\text{O}_3 + 4\text{H}_2\text{O}$
78. $\text{Fe}(\text{OH})_2 + \text{H}_2\text{SO}_4 \text{ (loãng)} \longrightarrow \text{FeSO}_4 + 2\text{H}_2\text{O}$
79. $2\text{Fe}(\text{OH})_2 + 4\text{H}_2\text{SO}_4 \text{ (đặc)} \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2\uparrow + 6\text{H}_2\text{O}$
80. $3\text{Fe}(\text{OH})_2 + 10\text{HNO}_3 \text{ (loãng)} \xrightarrow{t^\circ} 3\text{Fe}(\text{NO}_3)_3 + \text{NO}\uparrow + 8\text{H}_2\text{O}$
81. $\text{FeCl}_2 + 2\text{NaOH} \longrightarrow \text{Fe}(\text{OH})_2\downarrow + 2\text{NaCl}$
82. $2\text{FeCl}_2 + \text{Cl}_2 \longrightarrow 2\text{FeCl}_3$

83. $10\text{FeSO}_4 + 2\text{KMnO}_4 + 8\text{H}_2\text{SO}_4 \longrightarrow 5\text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 8\text{H}_2\text{O}$
84. $\text{Fe}_2\text{O}_3 + 6\text{HNO}_3 \longrightarrow 2\text{Fe}(\text{NO}_3)_3 + 3\text{H}_2\text{O}$
85. $\text{Fe}_2\text{O}_3 + 2\text{Al} \xrightarrow{r} \text{Al}_2\text{O}_3 + \text{Fe}$
86. $\text{Fe}_2\text{O}_3 + 3\text{CO} \xrightarrow{r} 2\text{Fe} + 3\text{CO}_2$
87. $2\text{Fe}(\text{OH})_3 \xrightarrow{r} \text{Fe}_2\text{O}_3 + 3\text{H}_2\text{O}$
88. $2\text{Fe}(\text{OH})_3 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$
89. $\text{FeCl}_3 + 3\text{NaOH} \longrightarrow \text{Fe}(\text{OH})_3 + 3\text{NaCl}$
90. $\text{Cu} + 2\text{FeCl}_3 \longrightarrow \text{CuCl}_2 + 2\text{FeCl}_2$
91. $2\text{FeCl}_3 + 2\text{KI} \longrightarrow 2\text{FeCl}_2 + 2\text{KCl} + \text{I}_2$
92. $\text{FeCl}_3 + 3\text{KSCN} \rightleftharpoons \text{Fe}(\text{SCN})_3 + 3\text{KCl}$
93. $\text{Fe}^{2+} + 6\text{CN}^- \rightarrow [\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
94. $\text{Fe}^{3+} + 6\text{CN}^- \rightarrow [\text{Fe}(\text{CN})_6]^{3-} \rightarrow \text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
95. $3\text{Fe}_3\text{O}_4 + \text{CO} \rightarrow 2\text{Fe}_3\text{O}_4 + \text{CO}_2$
96. $\text{Fe}_3\text{O}_4 + \text{CO} \rightarrow 3\text{FeO} + \text{CO}_2$
97. $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
98. $3\text{Fe} + \text{C} \xrightarrow{w} \text{Fe}_3\text{C}$
99. $3\text{Fe} + 2\text{CO} \xrightarrow{w} \text{Fe}_3\text{C} + \text{CO}_2$
100. $\text{CaCO}_3 \xrightarrow{w} \text{CaO} + \text{CO}_2$
101. $\text{CaO} + \text{SiO}_2(\text{cat}) \xrightarrow{w} \text{CaSiO}_3(\text{xi})$
102. $\text{P}_2\text{O}_5 + 3\text{CaO} \xrightarrow{w} \text{Ca}_3(\text{PO}_4)_2$
103. $4\text{Al} + 3\text{O}_2 \xrightarrow{r} 2\text{Al}_2\text{O}_3 \quad (\Delta H = -2.1675, 7\text{kJ})$
104. $4\text{Al} + 3\text{C} \xrightarrow{r} \text{Al}_4\text{C}_3$
105. $2\text{Al} + 3\text{S} \xrightarrow{r} \text{Al}_2\text{S}_3$
106. $\text{Al} + \text{P} \xrightarrow{r} \text{AlP}$
107. $8\text{Al} + 3\text{Fe}_3\text{O}_4 \xrightarrow{r} 9\text{Fe} + 4\text{Al}_2\text{O}_3 + \text{Q}$
108. $2\text{Al} + \text{Cr}_2\text{O}_3 \xrightarrow{r} 2\text{Cr} + \text{Al}_2\text{O}_3 + \text{Q}$
109. $\text{Al} + 3\text{HCl} \rightarrow \text{AlCl}_3 + 3/2\text{H}_2$
110. $8\text{Al} + 30\text{HNO}_3 \xrightarrow{r} 8\text{Al}(\text{NO}_3)_3 + 3\text{N}_2\text{O} + 15\text{H}_2\text{O}$
111. $2\text{Al} + 6\text{H}_2\text{SO}_4 \text{ đđc} \xrightarrow{r} \text{Al}_2(\text{SO}_4)_3 + 3\text{SO}_2 + 6\text{H}_2\text{O}$
112. $2\text{Al} + 2\text{NaOH} + 6\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4] + 3\text{H}_2$
113. $2\text{Al} + 2\text{NaOH} + \text{H}_2\text{O} \rightarrow 2\text{NaAlO}_2 + 3\text{H}_2$
114. $\text{SiO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SiO}_3 + \text{H}_2\text{O}$
115. $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$
116. $\text{NaAlO}_2 + \text{CO}_2 + 2\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{NaHCO}_3$
117. $2\text{Al}(\text{OH})_3 \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$
118. $2\text{Al}_2\text{O}_3 \xrightarrow{\text{đnc}} 4\text{Al} + 3\text{O}_2$
119. $\text{Al}_2\text{O}_3 + 9\text{C} \xrightarrow{>2000^\circ\text{C}} \text{Al}_4\text{C}_3 + 6\text{CO}$
120. $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
121. $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$
122. $2\text{Al}(\text{OH})_3 \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$
123. $\text{Al}(\text{OH})_3 + 3\text{HCl} \rightarrow \text{AlCl}_3 + 3\text{H}_2\text{O}$
124. $\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAlO}_2 + 2\text{H}_2\text{O}$ hay $\text{Al}(\text{OH})_3 + \text{OH}^- \rightarrow [\text{Al}(\text{OH})_4]^-$
125. $\text{AlCl}_3 + \text{NaOH} \rightarrow \text{Al}(\text{OH})_3 \downarrow + \text{NaCl}$ nếu dư $\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAlO}_2 + 2\text{H}_2\text{O}$
126. $2\text{AlCl}_3 + 3\text{Na}_2\text{CO}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{Al}(\text{OH})_3 \downarrow + 6\text{NaCl} + 3\text{CO}_2 \uparrow$
127. $\text{AlCl}_3 + 3\text{NH}_3 + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 \downarrow + 3\text{NH}_4\text{Cl}$
128. $2\text{NaAlO}_2 + \text{CO}_2 + 3\text{H}_2\text{O} \rightarrow 2\text{Al}(\text{OH})_3 \downarrow + \text{Na}_2\text{CO}_3$
129. $\text{NaAlO}_2 + \text{HCl}$ vừa đủ $+ \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 \downarrow + \text{NaCl}$
130. $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O} \rightarrow \text{K}^+ + \text{Al}^{3+} + 2\text{SO}_4^{2-} + 12\text{H}_2\text{O}$

131. $\text{Zn} + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2 \uparrow$
132. $\text{Al} + \text{NaOH} + 3\text{H}_2\text{O} \rightarrow \text{Na}[\text{Al}(\text{OH})_4] + 3/2\text{H}_2 \uparrow$
133. $\text{Zn} + 2\text{NaOH} + 2\text{H}_2\text{O} \rightarrow \text{Na}_2[\text{Zn}(\text{OH})_4] + \text{H}_2 \uparrow$
134. $\text{ZnO} + 2\text{OH}^- \rightarrow \text{ZnO}_2^{2-} + \text{H}_2\text{O}$
135. $\text{Zn}(\text{OH})_2 + 2\text{OH}^- \rightarrow \text{ZnO}_2^{2-} + 2\text{H}_2\text{O}$
136. $\text{ZnO}_2^{2-} + 2\text{CO}_2 + 2\text{H}_2\text{O} \rightarrow \text{Zn}(\text{OH})_2 \downarrow + 2\text{HCO}_3^-$
137. $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HClO}$
138. $\text{Cl}_2 + 2\text{KOH} \xrightarrow{\text{t}^\circ \text{thấp}} \text{KCl} + \text{KClO} + \text{H}_2\text{O}$
139. $3\text{Cl}_2 + 6\text{KOH} \xrightarrow{>75^\circ\text{C}} 5\text{KCl} + \text{KClO}_3 + 3\text{H}_2\text{O}$
140. $2\text{Cl}_2 + 2\text{Ca}(\text{OH})_2 \text{ loãng} \rightarrow \text{CaCl}_2 + \text{Ca}(\text{OCl})_2 + 2\text{H}_2\text{O}$
141. $\text{Cl}_2 + \text{Ca}(\text{OH})_2 \text{ huyền phù} \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$
142. $\text{Cl}_2 + 2\text{NaBr} \rightarrow 2\text{NaCl} + \text{Br}_2$ $\text{Cl}_2 + 2\text{NaI} \rightarrow 2\text{NaCl} + \text{I}_2$
143. $\text{SO}_2 + \text{Cl}_2 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$
144. $6\text{FeSO}_4 + 3\text{Cl}_2 \rightarrow 2\text{Fe}_2(\text{SO}_4)_3 + 2\text{FeCl}_3$
145. $\text{H}_2\text{S} + 4\text{Cl}_2 + 4\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 8\text{HCl}$
146. $\text{MnO}_2 + 4\text{HCl} \text{ đặc} \xrightarrow{\text{t}^\circ} \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$
147. $2\text{KMnO}_4 + 16\text{HCl} \rightarrow 2\text{KCl} + 2\text{MnCl}_2 + 5\text{Cl}_2 + 8\text{H}_2\text{O}$
148. $2\text{NaCl} + 2\text{H}_2\text{O} \xrightarrow[\text{màng}]{\text{điện}} 2\text{NaOH} + \text{Cl}_2 \uparrow + \text{H}_2 \uparrow$
149. $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$
150. $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$
151. $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
152. $\text{FeS} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2\text{S} \uparrow$
153. $\text{Na}_2\text{SO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{SO}_2 \uparrow + \text{H}_2\text{O}$
154. $\text{AgNO}_3 + \text{HCl} \rightarrow \text{AgCl} \downarrow + \text{HNO}_3$
155. $\text{NaCl} \text{ tinh thể} + \text{H}_2\text{SO}_4 \text{ đặc} \xrightarrow{\text{t}^\circ} \text{NaHSO}_4 + \text{HCl} \uparrow$
156. $2\text{NaCl} \text{ tinh thể} + \text{H}_2\text{SO}_4 \text{ đặc} \xrightarrow{\text{t}^\circ} 2\text{Na}_2\text{SO}_4 + \text{HCl} \uparrow$
157. $\text{Cl}_2 + 2\text{KOH} \rightarrow \text{KCl} + \text{KClO} + \text{H}_2\text{O}$
158. $2\text{H}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_2 + 2\text{H}_2\text{O}$
159. $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$
160. $\text{O}_3 + 2\text{KI} + \text{H}_2\text{O} \rightarrow \text{O}_2 + 2\text{KOH} + \text{I}_2$
161. $\text{H}_2\text{S} + 2\text{NaOH} \rightarrow \text{Na}_2\text{S} + 2\text{H}_2\text{O}$
162. $\text{H}_2\text{S} + \text{NaOH} \rightarrow \text{NaHS} + \text{H}_2\text{O}$
163. $\text{H}_2\text{S} + \text{Cu}(\text{NO}_3)_2 \rightarrow \text{CuS} \downarrow_{\text{đen}} + 2\text{HNO}_3$
164. $2\text{H}_2\text{S} + 3\text{O}_2 \xrightarrow{\text{t}^\circ} 2\text{SO}_2 + 2\text{H}_2\text{O}$
165. $2\text{H}_2\text{S} + \text{O}_2 \text{ oxy hoá chậm} \xrightarrow{\text{t}^\circ} 2\text{S} + 2\text{H}_2\text{O}$
166. $\text{H}_2\text{S} + 4\text{Cl}_2 + 4\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 8\text{HCl}$
167. $2\text{SO}_2 + \text{O}_2 \xrightleftharpoons[450-500^\circ\text{C}]{\text{V}_2\text{O}_5} 2\text{SO}_3$
168. $\text{SO}_2 + \text{Br}_2 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 2\text{HBr}$
169. $\text{SO}_2 + \text{Cl}_2 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$
170. $2\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$
171. $\text{Na}_2\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{SO}_2 \uparrow + \text{H}_2\text{O}$
172. $\text{Cu} + 2\text{H}_2\text{SO}_4 \text{ đặc} \xrightarrow{\text{t}^\circ} \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$

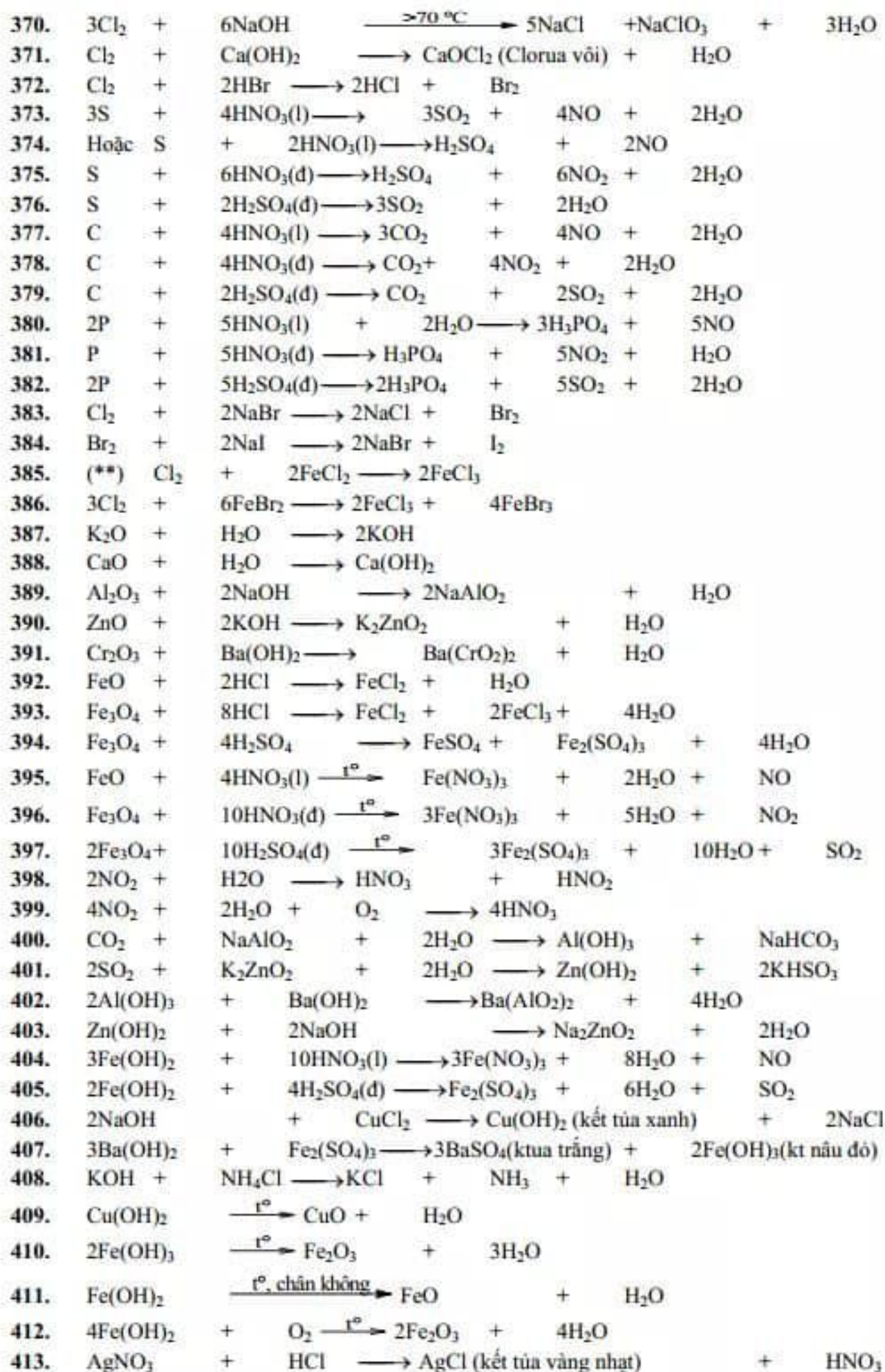
173. $\text{MgCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
174. $\text{Na}_2\text{CO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
175. $\text{H}_2\text{SO}_4 \text{ đặc} + \text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
176. $2\text{Fe} + 6\text{H}_2\text{SO}_4 \text{ đặc} \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + 3\text{SO}_2 + 6\text{H}_2\text{O}$
177. $2\text{FeO} + 4\text{H}_2\text{SO}_4 \text{ đặc} \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 4\text{H}_2\text{O}$
178. $2\text{FeCO}_3 + 4\text{H}_2\text{SO}_4 \text{ đặc} \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 2\text{CO}_2 + 4\text{H}_2\text{O}$
179. $2\text{Fe}_3\text{O}_4 + 10\text{H}_2\text{SO}_4 \text{ đặc} \rightarrow 3\text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 10\text{H}_2\text{O}$
180. $2\text{FeSO}_4 + 2\text{H}_2\text{SO}_4 \text{ đặc} \rightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 2\text{H}_2\text{O}$
181. $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$
182. $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$
183. $4\text{NH}_3 + 3\text{O}_2 \xrightarrow{t^\circ} 2\text{N}_2 + 6\text{H}_2\text{O}$
184. $\text{AlCl}_3 + 3\text{NH}_3 + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 \downarrow + 3\text{NH}_4\text{Cl}$
185. $4\text{NH}_3 + 5\text{O}_2 \xrightarrow[\text{Pt}]{450-500^\circ\text{C}} 4\text{NO} + 6\text{H}_2\text{O}$
186. $\text{Cu}(\text{OH})_2 \downarrow + 4\text{NH}_3 \text{ (dd)} \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} \text{ (dd)} + 2\text{OH}^- \text{ (dd)}$
187. $\text{N}_2 + 3\text{H}_2 \xrightleftharpoons[200-300 \text{ (atm), Fe}]{450-500^\circ\text{C}} 2\text{NH}_3$
188. $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NaCl} + \text{NH}_3 \uparrow + \text{H}_2\text{O}$
189. $\text{NH}_4\text{HCO}_3 \xrightarrow{t^\circ} \text{NH}_3 \uparrow + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
190. $\text{NH}_4\text{NO}_2 \xrightarrow{t^\circ} \text{N}_2 + 2\text{H}_2\text{O}$ Hoặc: $\text{NH}_4\text{NO}_3 \xrightarrow{t^\circ} \text{N}_2\text{O} + 2\text{H}_2\text{O}$
191. $\text{HNO}_3 + \text{NaOH} \rightarrow \text{NaNO}_3 + \text{H}_2\text{O}$
192. $2\text{HNO}_3 + \text{Mg}(\text{OH})_2 \rightarrow \text{Mg}(\text{NO}_3)_2 + 2\text{H}_2\text{O}$
193. $\text{Fe}_2\text{O}_3 + 6\text{HNO}_3 \rightarrow 2\text{Fe}(\text{NO}_3)_3 + 3\text{H}_2\text{O}$
194. $\text{CuO} + 2\text{HNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + \text{H}_2\text{O}$
195. $\text{Fe} + 4\text{HNO}_3 \text{ loãng} \rightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO} \uparrow + 2\text{H}_2\text{O}$
196. $10\text{Al} + 36\text{HNO}_3 \rightarrow 10\text{Al}(\text{NO}_3)_3 + 3\text{N}_2 \uparrow + 18\text{H}_2\text{O}$
197. $8\text{Al} + 30\text{HNO}_3 \rightarrow 8\text{Al}(\text{NO}_3)_3 + 3\text{N}_2\text{O} \uparrow + 15\text{H}_2\text{O}$
198. $\text{KNO}_3 \rightarrow \text{K}^+ + \text{NO}_3^-$ và $\text{H}_2\text{SO}_4 \rightarrow 2\text{H}^+ + \text{SO}_4^{2-}$
199. $4\text{Mg} + 10\text{HNO}_3 \rightarrow 4\text{Mg}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + 3\text{H}_2\text{O}$
200. $\text{KNO}_3 \rightarrow \text{K}^+ + \text{NO}_3^-$ và $\text{H}_2\text{SO}_4 \rightarrow 2\text{H}^+ + \text{SO}_4^{2-}$
201. $3\text{FeO} + 10\text{HNO}_3 \rightarrow 3\text{Fe}(\text{NO}_3)_3 + \text{NO} + 5\text{H}_2\text{O}$
202. $\text{Fe}_3\text{O}_4 + 10\text{HNO}_3 \rightarrow 3\text{Fe}(\text{NO}_3)_3 + \text{NO}_2 + 5\text{H}_2\text{O}$
203. $\text{FeCO}_3 + 4\text{HNO}_3 \rightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2 + \text{CO}_2 + 2\text{H}_2\text{O}$
204. $3\text{Fe}^{2+} + \text{NO}_3^- + 4\text{H}^+ \rightarrow 3\text{Fe}^{3+} + \text{NO} + 2\text{H}_2\text{O}$
205. $\text{FeS}_2 + 18\text{HNO}_3 \rightarrow \text{Fe}(\text{NO}_3)_3 + 2\text{H}_2\text{SO}_4 + 15\text{NO}_2 + 7\text{H}_2\text{O}$
206. $\text{C} + 4\text{HNO}_3 \rightarrow \text{CO}_2 + 4\text{NO}_2 + 2\text{H}_2\text{O}$
207. $\text{S} + 6\text{HNO}_3 \rightarrow \text{H}_2\text{SO}_4 + 6\text{NO}_2 + 2\text{H}_2\text{O}$
208. $4\text{NH}_3 + 5\text{O}_2 \xrightarrow[\text{Pt}]{450^\circ\text{C}} 4\text{NO} + 6\text{H}_2\text{O}$
209. $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$;
210. $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$
211. $2\text{KNO}_3 \xrightarrow{t^\circ} 2\text{KNO}_2 + \text{O}_2$
212. $2\text{Pb}(\text{NO}_3)_2 \xrightarrow{t^\circ} 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$
213. $2\text{Cu}(\text{NO}_3)_2 \xrightarrow{t^\circ} 2\text{CuO} + 4\text{NO}_2 + \text{O}_2$
214. $2\text{AgNO}_3 \xrightarrow{t^\circ} 2\text{Ag} + 2\text{NO}_2 + \text{O}_2$
215. $3\text{C} + 2\text{KClO}_3 \xrightarrow{t^\circ} 2\text{KCl} + 3\text{CO}_2$
216. $\text{C} + 2\text{CuO} \xrightarrow{t^\circ} 2\text{Cu} + \text{CO}_2$

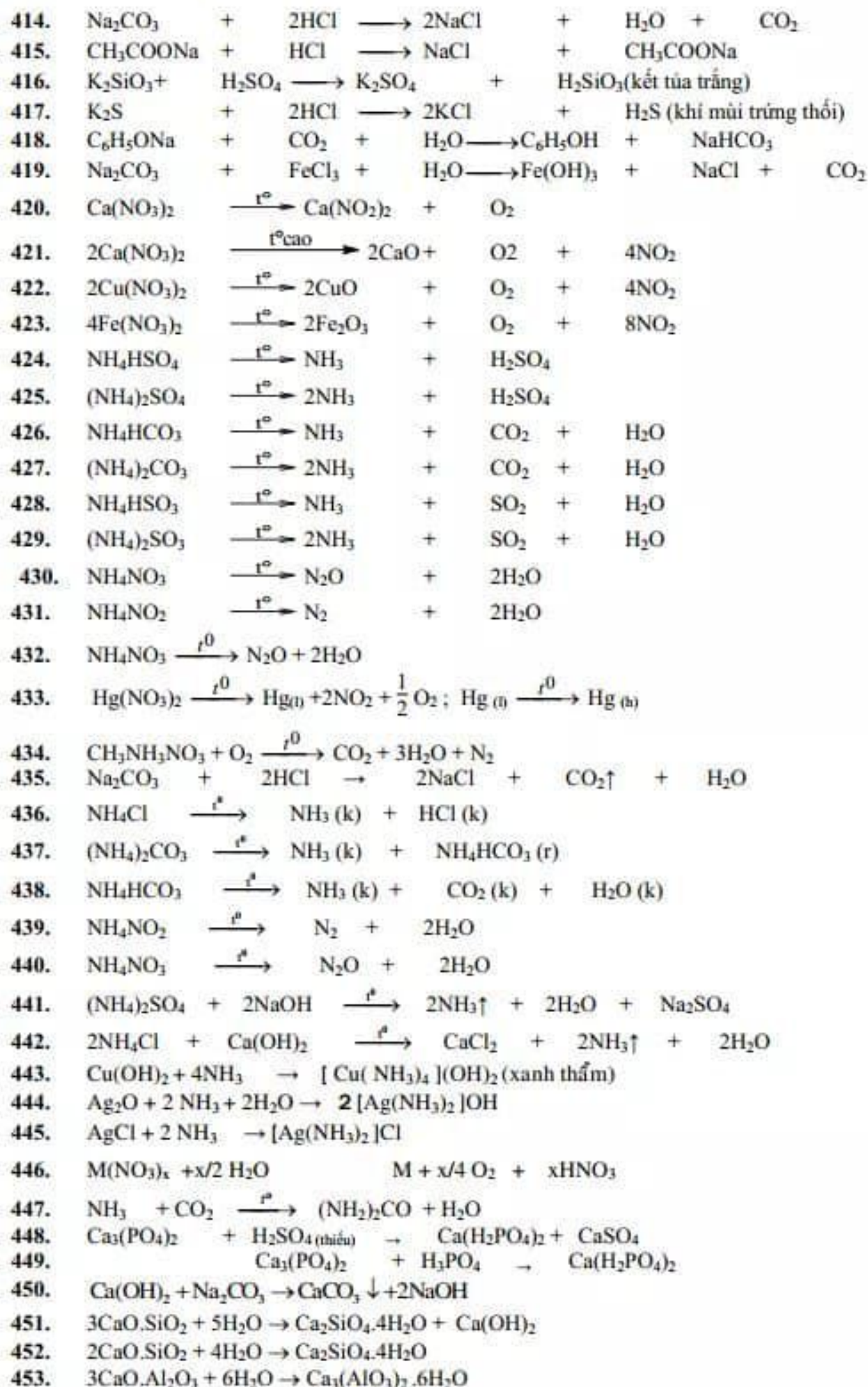
217. $C + ZnO \xrightarrow{t^o} Zn + CO$
218. $Ca + 2 C \xrightarrow{t^o} CaC_2$
219. $4 Al + 3 C \xrightarrow{t^o} Al_4C_3$
220. $Al_4C_3 + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_4 \uparrow$
221. $PdCl_2 + H_2O + CO \rightarrow Pd + 2 HCl + CO_2$
222. $CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$
223. $CO_2 + 2 NaOH \rightarrow Na_2CO_3 + H_2O$
224. $Na_2CO_3 + CO_2 + H_2O \rightarrow NaHCO_3$
225. $CaCO_3 + 2 HCl \rightarrow CaCl_2 + CO_2 \uparrow + H_2O$
226. $Si + O_2 \xrightarrow{t^o} SiO_2$
227. $Si + 2NaOH + H_2O \xrightarrow{t^o} Na_2SiO_3 + 2H_2$
228. $SiO_2 + CaO \xrightarrow{t^o} CaSiO_3 \text{ (canxi silicat)}$
229. $SiO_2 + 2NaOH \xrightarrow{t^o} Na_2SiO_3 + H_2O$
230. $SiO_2 + K_2CO_3 \xrightarrow{t^o} K_2SiO_3 + CO_2 \uparrow$
231. $2HCl + Na_2SiO_3 \rightarrow H_2SiO_3 + 2NaCl$
232. $Al + NaOH + H_2O \rightarrow NaAlO_2 + 3/2H_2$
233. $AlCl_3 + 4NaOH \rightarrow NaAlO_2 + 3NaCl + 2H_2O$
234. $NaHCO_3 + Ca(OH)_2 \rightarrow CaCO_3 + NaOH + H_2O$
235. $NaAlO_2 + 4HCl \rightarrow AlCl_3 + NaCl + 2H_2O$
236. $Cl_2 + 6KOH \longrightarrow 5KCl + KClO_3 + 3H_2O$
237. $Cl_2 + 2KOH \rightarrow KCl + KClO + 2H_2O$
238. $2AlCl_3 + 3Na_2CO_3 + 3H_2O \longrightarrow 2Al(OH)_3 + NaCl + 3CO_2$
239. $Al_2(SO_4)_3 + 3Na_2CO_3 + 3H_2O \longrightarrow 2Al(OH)_3 + 3Na_2SO_4 + 3CO_2$
240. $2AlCl_3 + 3Na_2S + 6H_2O \longrightarrow 2Al(OH)_3 + 6NaCl + 3H_2S$
241. $NaAlO_2 + NH_4Cl + H_2O \longrightarrow Al(OH)_3 + NaCl + NH_3$
242. $2FeCl_3 + 3Na_2CO_3 + 3H_2O \longrightarrow 2Fe(OH)_3 + 6NaCl + 3CO_2$
243. $2FeCl_3 + 3Na_2S + 6H_2O \longrightarrow 2Fe(OH)_3 + 6NaCl + 3H_2S$
244. $3KNO_3 + 5KOH + 8Al + 2H_2O \longrightarrow 8KAlO_2 + 3NH_3$
245. $(KOH + Al + H_2O \longrightarrow KAlO_2 + 3/2H_2)$
246. $2KNO_3 + 4H_2SO_4 + 3Cu \longrightarrow 3CuSO_4 + K_2SO_4 + 2NO + 4H_2O$
247. $(3Cu + 2NO_3^- + 8H^+ \longrightarrow 3Cu^{2+} + 2NO + 4H_2O)$
248. $H_2SO_4 \text{ đặc} + KL \longrightarrow MSO_4^{2-} + \{SO_2, S, H_2S\} + H_2O$
249. $2H_2SO_4 \text{ đặc} + S \longrightarrow 3SO_2 + 2H_2O$
250. $2H_2SO_4 \text{ đặc} + C \longrightarrow CO_2 + 2SO_2 + 2H_2O$
251. $H_2SO_4 \text{ đặc} + H_2S \longrightarrow SO_2 + S + 2H_2O$
252. $4H_2SO_4 \text{ đặc} + 2FeO \longrightarrow Fe_2(SO_4)_3 + SO_2 + 4H_2O$
253. $10H_2SO_4 \text{ đặc} + 2Fe_3O_4 \longrightarrow 3Fe_2(SO_4)_3 + SO_2 + 10H_2O$

254. $4\text{H}_2\text{SO}_4 \text{ đặc} + 2\text{Fe}(\text{OH})_2 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 6\text{H}_2\text{O}$
255. $14\text{H}_2\text{SO}_4 \text{ đặc} + 2\text{FeS}_2 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 15\text{SO}_2 + 14\text{H}_2\text{O}$
256. $2\text{H}_2\text{SO}_4 \text{ đặc} + 2\text{NaBr} \longrightarrow \text{Na}_2\text{SO}_4 + \text{SO}_2 + \text{Br}_2 + 2\text{H}_2\text{O}$
257. $5\text{H}_2\text{SO}_4 \text{ đặc} + 8\text{NaI} \longrightarrow 4\text{Na}_2\text{SO}_4 + \text{H}_2\text{S} + 4\text{I}_2 + 4\text{H}_2\text{O}$
258. $\text{HNO}_3 + \text{KI} \longrightarrow \text{MNO}_3^- + [\text{NO}_2, \text{NO}, \text{N}_2\text{O}, \text{N}_2, \text{NH}_4\text{NO}_3] + \text{H}_2\text{O}$
259. $\text{HNO}_3 + \text{Au} + 3\text{HCl} \longrightarrow \text{AuCl}_3 + \text{NO} + 2\text{H}_2\text{O}$
260. $6\text{HNO}_3 \text{ đặc} + \text{S} \xrightarrow{\text{nhiệt độ}} \text{H}_2\text{SO}_4 + 6\text{NO}_2 + 2\text{H}_2\text{O}$
261. $5\text{HNO}_3 \text{ đặc} + \text{P} \xrightarrow{\text{nhiệt độ}} \text{H}_3\text{PO}_4 + 5\text{NO}_2 + \text{H}_2\text{O}$
262. $5\text{HNO}_3 \text{ loãng} + 3\text{P} + 2\text{H}_2\text{O} \longrightarrow 3\text{H}_3\text{PO}_4 + 5\text{NO}$
263. $4\text{HNO}_3 \text{ đặc} + \text{C} \xrightarrow{\text{nhiệt độ}} \text{CO}_2 + 4\text{NO}_2 + 2\text{H}_2\text{O}$
264. $10\text{HNO}_3 + 3\text{I}_2 \xrightarrow{\text{nhiệt độ}} 6\text{HI} + 10\text{NO} + 2\text{H}_2\text{O}$
265. $4\text{HNO}_3 \text{ đặc} + \text{FeO} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2 + 2\text{H}_2\text{O}$
266. $10\text{HNO}_3 \text{ loãng} + 3\text{FeO} \longrightarrow 3\text{Fe}(\text{NO}_3)_3 + \text{NO} + 5\text{H}_2\text{O}$
267. $10\text{HNO}_3 \text{ đặc} + \text{Fe}_3\text{O}_4 \longrightarrow 3\text{Fe}(\text{NO}_3)_3 + \text{NO}_2 + 5\text{H}_2\text{O}$
268. $28\text{HNO}_3 \text{ loãng} + 3\text{Fe}_3\text{O}_4 \longrightarrow 9\text{Fe}(\text{NO}_3)_3 + \text{NO} + 14\text{H}_2\text{O}$
269. $4\text{HNO}_3 \text{ đặc} + \text{Fe}(\text{OH})_2 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO}_2 + 3\text{H}_2\text{O}$
270. $10\text{HNO}_3 \text{ loãng} + 3\text{Fe}(\text{OH})_2 \longrightarrow 3\text{Fe}(\text{NO}_3)_3 + \text{NO} + 8\text{H}_2\text{O}$
271. $48\text{HNO}_3 \text{ đặc} + 3\text{FeS}_2 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{SO}_4 + 45\text{NO}_2 + 21\text{H}_2\text{O}$
272. $18\text{HNO}_3 \text{ loãng} + 3\text{FeS}_2 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{SO}_4 + 15\text{NO} + 6\text{H}_2\text{O}$
273. *Lưu ý:* Hai phản ứng trên nên viết dưới dạng phản ứng ion rút gọn:
274. $14\text{H}^+ + 15\text{NO}_3^- + \text{FeS}_2 \longrightarrow \text{Fe}^{3+} + 15\text{NO}_2 + 2\text{SO}_4^{2-} + 7\text{H}_2\text{O}$
275. $4\text{H}^+ + 5\text{NO}_3^- + \text{FeS}_2 \longrightarrow \text{Fe}^{3+} + 5\text{NO} + 2\text{SO}_4^{2-} + 2\text{H}_2\text{O}$
276. $30\text{HNO}_3 \text{ đặc} + 3\text{FeS} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{Fe}_2(\text{SO}_4)_3 + 27\text{NO}_2 + 15\text{H}_2\text{O}$
277. $12\text{HNO}_3 \text{ loãng} + 3\text{FeS} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{Fe}_2(\text{SO}_4)_3 + 9\text{NO} + 6\text{H}_2\text{O}$
278. $10\text{H}^+ + 9\text{NO}_3^- + \text{FeS} \longrightarrow \text{Fe}^{3+} + 9\text{NO}_2 + \text{SO}_4^{2-} + 5\text{H}_2\text{O}$
279. $4\text{H}^+ + 3\text{NO}_3^- + \text{FeS} \longrightarrow \text{Fe}^{3+} + 3\text{NO} + \text{SO}_4^{2-} + 2\text{H}_2\text{O}$
280. $4\text{HNO}_3 \text{ đặc} + \text{FeCO}_3 \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{CO}_2 + \text{NO}_2 + 2\text{H}_2\text{O}$
281. $10\text{HNO}_3 \text{ loãng} + 3\text{FeCO}_3 \longrightarrow 3\text{Fe}(\text{NO}_3)_3 + 3\text{CO}_2 + \text{NO} + 5\text{H}_2\text{O}$
282. $\text{O}_3 + 2\text{KI} + \text{H}_2\text{O} \longrightarrow 2\text{KOH} + \text{I}_2 + \text{O}_2$
283. $\text{O}_2 + \text{S} \xrightarrow{\text{nhiệt độ}} \text{SO}_2$
284. $\text{O}_2 + 2\text{SO}_2 \xrightarrow[450^\circ\text{C}]{\text{V}_2\text{O}_5} 2\text{SO}_3$
285. $\text{O}_2 + 2\text{H}_2\text{S} \longrightarrow 2\text{S} + 2\text{H}_2\text{O}$
286. $3\text{O}_2 + 2\text{H}_2\text{S} \longrightarrow 2\text{SO}_2 + 2\text{H}_2\text{O}$
287. $3\text{O}_2 + 4\text{NH}_3 \xrightarrow{\text{đốt cháy}} 2\text{N}_2 + 6\text{H}_2\text{O}$
288. $5\text{O}_2 + 4\text{NH}_3 \xrightarrow[850^\circ\text{C}]{\text{Pt}} 4\text{NO} + 6\text{H}_2\text{O}$

289. $1/2\text{O}_2 + \text{H}_2\text{O} + 2\text{Fe}(\text{OH})_2 \longrightarrow 2\text{Fe}(\text{OH})_3$
290. $1/2\text{O}_2 + 2\text{HCl} + 2\text{FeCl}_2 \longrightarrow 2\text{FeCl}_3 + \text{H}_2\text{O}$
291. $3/2\text{O}_2 + 3\text{H}_2\text{O} + 6\text{FeSO}_4 \longrightarrow 2\text{Fe}(\text{OH})_3 + 2\text{Fe}_2(\text{SO}_4)_3$
292. $3\text{F}_2 + 4\text{NH}_3 \longrightarrow 3\text{NH}_4\text{F} + \text{NF}_3$
293. $\text{Cl}_2 + 2\text{FeCl}_2 \longrightarrow 2\text{FeCl}_3$
294. $3/2\text{Cl}_2 + 3\text{FeSO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + \text{FeCl}_3$
295. $3\text{Cl}_2 + 8\text{NH}_3 \longrightarrow 6\text{NH}_4\text{Cl} + \text{N}_2$
296. $\text{Cl}_2 + 2\text{H}_2\text{O} + \text{SO}_2 \longrightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$
297. $4\text{Cl}_2 + 4\text{H}_2\text{O} + \text{H}_2\text{S} \longrightarrow \text{H}_2\text{SO}_4 + 8\text{HCl}$
298. $\text{Cl}_2 + \text{H}_2\text{O} + \text{H}_2\text{SO}_3 \longrightarrow \text{H}_2\text{SO}_4 + 2\text{HCl}$
299. $2\text{FeCl}_3 + 2\text{KI} \longrightarrow 2\text{FeCl}_2 + \text{I}_2 + 2\text{KCl}$
300. $\text{FeCl}_3 + 2\text{HI} \longrightarrow \text{FeCl}_2 + \text{I}_2 + 2\text{HCl}$
301. $2\text{FeCl}_3 + \text{Na}_2\text{S} \longrightarrow 2\text{FeCl}_2 + \text{S} + 2\text{NaCl}$
302. $2\text{FeCl}_3 + \text{H}_2\text{S} \longrightarrow 2\text{FeCl}_2 + \text{S} + 2\text{HCl}$
303. $4\text{KNO}_3 + \text{C} \xrightarrow{\text{nhiệt độ}} 2\text{K}_2\text{O} + \text{CO}_2 + 4\text{NO}_2$
304. $2\text{KNO}_3 + \text{S} + 3\text{C} \xrightarrow{\text{nhiệt độ}} \text{K}_2\text{S} + 3\text{CO}_2 + \text{N}_2$
305. $2\text{KClO}_3 + 3\text{C} \xrightarrow{\text{nhiệt độ}} 2\text{KCl} + 3\text{CO}_2$
306. $2\text{KClO}_3 + 3\text{S} \xrightarrow{\text{nhiệt độ}} 2\text{KCl} + 3\text{SO}_2$
307. $5\text{KClO}_3 + 6\text{P} \xrightarrow{\text{nhiệt độ}} 5\text{KCl} + 3\text{P}_2\text{O}_5$
308. $2\text{NH}_3 + 3\text{CuO} \xrightarrow{\text{nhiệt độ}} \text{N}_2 + 3\text{Cu} + 3\text{H}_2\text{O}$
309. $2\text{H}_2\text{S} + \text{SO}_2 \longrightarrow 3\text{S} + 2\text{H}_2\text{O}$
310. $\text{NH}_4\text{NO}_2 \xrightarrow{\text{nhiệt phân}} \text{N}_2 + 2\text{H}_2\text{O}$
311. $\text{NH}_4\text{NO}_3 \xrightarrow{>200^\circ\text{C}} \text{N}_2 + 1/2\text{O}_2 + 2\text{H}_2\text{O}$
312. $\text{NH}_4\text{NO}_3 \xrightarrow{<200^\circ\text{C}} \text{N}_2\text{O} + 2\text{H}_2\text{O} \text{ (phản ứng nổ)}$
313. $3(\text{NH}_4)_2\text{SO}_4 \xrightarrow{\text{nhiệt phân}} \text{N}_2 + 4\text{NH}_3 + 3\text{SO}_2 + 6\text{H}_2\text{O}$
314. $(\text{NH}_4)_3\text{PO}_4 \xrightarrow{\text{nhiệt phân}} 3\text{NH}_3 + \text{HPO}_3 + \text{H}_2\text{O}$
315. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\text{nhiệt phân}} \text{N}_2 + \text{Cr}_2\text{O}_3 + 4\text{H}_2\text{O}$
316. $3\text{Cu} + 8\text{HNO}_3 \longrightarrow 2\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$
317. $2\text{HCO}_3^- \xrightarrow{t^\circ} \text{CO}_2\uparrow + \text{CO}_3^{2-} + \text{H}_2\text{O}$
318. $2\text{HCO}_3^- \xrightarrow{t^\circ} \text{CO}_2\uparrow + \text{CO}_3^{2-} + \text{H}_2\text{O}$
319. $\text{Mg}^{2+} + 2\text{OH}^- \rightarrow \text{Mg}(\text{OH})_2\downarrow$
320. $\text{Al}^{3+} + 3\text{OH}^- \rightarrow \text{Al}(\text{OH})_3\downarrow$
321. $\text{Al}(\text{OH})_3 + \text{OH}^- \rightarrow \text{AlO}_2^- + 2\text{H}_2\text{O}$
322. $3\text{H}_2\text{S} + 2\text{KMnO}_4 \rightarrow 2\text{MnO}_2 + 3\text{S}\downarrow + 2\text{KOH} + 2\text{H}_2\text{O}$
323. $5\text{H}_2\text{S} + 2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \rightarrow 2\text{MnSO}_4 + 5\text{S}\downarrow + \text{K}_2\text{SO}_4 + 8\text{H}_2\text{O}$
324. $\text{H}_2\text{S} + 2\text{FeCl}_3 \rightarrow 2\text{FeCl}_2 + \text{S}\downarrow + 2\text{HCl}$
325. $\text{SO}_2 + 2\text{KMnO}_4 + 2\text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4 + 2\text{MnSO}_4 + \text{K}_2\text{SO}_4$

326. $\text{Cu}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{CuCl}_2 + 2\text{H}_2\text{O}$
327. $\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{FeSO}_4 + 7\text{H}_2\text{SO}_4 \rightarrow 3\text{Fe}_2(\text{SO}_4)_3 + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$
328. $2\text{NaCrO}_2 + 3\text{Br}_2 + 8\text{NaOH} \rightarrow 2\text{Na}_2\text{CrO}_4 + 6\text{NaBr} + 4\text{H}_2\text{O}$
329. $2\text{CrCl}_3 + \text{Zn} \rightarrow 2\text{CrCl}_2 + \text{ZnCl}_2$
330. $\text{Cr}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaCrO}_2 + 2\text{H}_2\text{O}$
331. $\text{AlCl}_3 + 3\text{NH}_3 + 3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 \downarrow + 3\text{NH}_4\text{Cl}$
332. $\text{Ca}(\text{HCO}_3)_2 \xrightarrow{r} \text{CaCO}_3 \downarrow + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
333. $\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{CaCO}_3 \downarrow + 2\text{H}_2\text{O}$
334. $\text{Ca}(\text{HCO}_3)_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 \downarrow + 2\text{NaHCO}_3$
335. $\text{CaSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 \downarrow + \text{Na}_2\text{SO}_4$
336. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{r} \text{CaSO}_4 \cdot \text{H}_2\text{O}$
337. $4\text{Mg} + 10\text{HNO}_3 \text{ (loãng)} \rightarrow 4\text{Mg}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + 3\text{H}_2\text{O}$
338. $4\text{Mg} + 5\text{H}_2\text{SO}_4 \text{ (đặc)} \rightarrow 4\text{MgSO}_4 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$
339. $2\text{NaHCO}_3 \xrightarrow{r} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
340. $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$
341. $\text{NaHCO}_3 + \text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$
342. $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$
343. $4\text{AgNO}_3 + 2\text{H}_2\text{O} \xrightarrow{\text{đốt}} 4\text{Ag} + \text{O}_2 + 4\text{HNO}_3$
344. $\text{CuSO}_4 + 2\text{H}_2\text{O} \xrightarrow{\text{đốt}} 2\text{Cu} + 2\text{H}_2\text{SO}_4 + \text{O}_2$
345. $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$
346. $3\text{Fe} + 4\text{H}_2\text{O} \xrightarrow{700^\circ\text{C}} \text{Fe}_3\text{O}_4 + 4\text{H}_2$
347. $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$
348. $2\text{Na} + \text{O}_2 \text{ (đur)} \rightarrow \text{Na}_2\text{O}_2 \text{ (Natri peoxit)}$
349. $3\text{Ca} + \text{N}_2 \xrightarrow{t^\circ} \text{Ca}_3\text{N}_2 \text{ (Canxi nitrua)}$
350. $2\text{Na} + \text{CuO} \xrightarrow{t^\circ, \text{Chân không}} \text{Cu} + \text{Na}_2\text{O}$
351. $\text{Al} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{NaAlO}_2 + 3/2 \text{H}_2$
352. $\text{Zn} + 2\text{NaOH} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$
353. $2\text{Al} + \text{Ba}(\text{OH})_2 + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{AlO}_2)_2 + 3\text{H}_2$
354. $2\text{Al} + 3\text{H}_2\text{SO}_4 \text{ (loãng)} \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2$
355. $\text{Al} + \text{HNO}_3 \text{ (rất loãng)} \xrightarrow{t^\circ} \text{Al}(\text{NO}_3)_3 + \text{H}_2\text{O} + \text{NH}_4\text{NO}_3$
356. $\text{Fe} + \text{CuSO}_4 \rightarrow \text{FeSO}_4 + \text{Cu}$
357. $\text{Fe} + 2\text{FeCl}_3 \rightarrow 3\text{FeCl}_2$
358. $\text{Cu} + 2\text{FeCl}_3 \rightarrow 2\text{FeCl}_2 + \text{CuCl}_2$
359. $\text{Fe}(\text{NO}_3)_2 + \text{AgNO}_3 \rightarrow \text{Fe}(\text{NO}_3)_3 + \text{Ag} \downarrow$
360. $2\text{KClO}_3 + 12\text{HCl} \xrightarrow{t^\circ} 2\text{KCl} + 5\text{Cl}_2 + 6\text{H}_2\text{O}$
361. $\text{MnO}_2 + 4\text{HCl} \xrightarrow{t^\circ} \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$
362. $2\text{KMnO}_4 + 14\text{HCl} \xrightarrow{t^\circ} 2\text{MnCl}_2 + 2\text{KCl} + 5\text{Cl}_2 + 8\text{H}_2\text{O}$
363. $\text{H}_2\text{SO}_4 \text{ (đặc)} + 2\text{HCl} \xrightarrow{t^\circ} \text{SO}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
364. $3\text{C} + \text{Fe}_2\text{O}_3 \xrightarrow{t^\circ, \text{CNO}} 2\text{Fe} + 3\text{CO}$
365. $\text{SO}_2 + \text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + \text{HCl}$
366. $\text{SO}_2 + \text{Br}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + \text{HBr}$
367. $\text{SO}_2 + \text{I}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + \text{HI}$
368. $\text{X}_2 + 2\text{NaOH} \rightarrow \text{NaX} + \text{NaXO} + \text{H}_2\text{O}$
369. $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O} \text{ (Nước Javen)}$





454. $\text{H}_2\text{SiO}_3 \xrightarrow{t^\circ} \text{SiO}_2 + 2\text{H}_2\text{O}$
455. $\text{Na}_2\text{SiO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{SiO}_3 \downarrow$
456. $\text{SiO}_2 + 2\text{NaOH} \xrightarrow{t^\circ} \text{Na}_2\text{SiO}_3 + \text{H}_2\text{O}$
457. $\text{SiO}_2 + 2\text{Na}_2\text{CO}_3 \xrightarrow{t^\circ} \text{Na}_2\text{SiO}_3 + \text{CO}_2$
458. $\text{SiO}_2 + 4\text{HF} \rightarrow \text{SiF}_4 + 2\text{H}_2\text{O}$
459. $\text{SiO}_2 + 2\text{Mg} \xrightarrow{t^\circ} 2\text{MgO} + \text{Si}$
460. $\text{SiO}_2 + 2\text{C}_{\text{than cốc}} \xrightarrow{t^\circ} 2\text{CO} + \text{Si}$
461. $\text{Si} + \text{F}_2 \rightarrow \text{SiF}_4$; $\text{Si} + \text{O}_2 \xrightarrow{t^\circ} \text{SiO}_2$
462. $\text{Si} + 2\text{NaOH} + \text{H}_2\text{O} \xrightarrow{t^\circ} \text{Na}_2\text{SiO}_3 + \text{H}_2$
463. $\text{Si} + 2\text{Mg} \xrightarrow{t^\circ} \text{Mg}_2\text{Si}$
464. $\text{NaAlO}_2 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Al(OH)}_3 \downarrow + \text{NaHCO}_3$
465. $\text{C}_6\text{H}_5\text{ONa} + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_5\text{OH} + \text{NaHCO}_3$
466. $\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow \text{NaHCO}_3 + \text{NaCl}$
467. $\text{HCl} + \text{NaHCO}_3 \rightarrow \text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$
468. $\text{CaCO}_3 \xrightarrow{t^\circ} \text{CaO} + \text{CO}_2$; gốc HCO_3 bị nhiệt phân
469. $2\text{NaHCO}_3 \xrightarrow{t^\circ} \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
470. $\text{Ca(HCO}_3)_2 \xrightarrow{t^\circ} \text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
471. $\text{C} + \text{O}_2 \xrightarrow{t^\circ} \text{CO}_2$; $\text{CO}_2 + \text{C} \xrightarrow{t^\circ} 2\text{CO}$; $\text{Fe}_2\text{O}_3 + 3\text{C} \xrightarrow{t^\circ} 2\text{Fe} + 3\text{CO}$
472. $\text{CaO} + 3\text{C} \xrightarrow{t^\circ} \text{CaC}_2 + \text{CO}$; $2\text{C} + \text{Ca} \xrightarrow{t^\circ} \text{CaC}_2$
473. $\text{FeO} + \text{CO} \xrightarrow{t^\circ} \text{Fe} + \text{CO}_2$
474. $\text{C} + \text{H}_2\text{O} \xrightarrow{t^\circ} \text{CO} + \text{H}_2$; $4\text{Al} + 3\text{C} \xrightarrow{t^\circ} \text{Al}_4\text{C}_3$
475. $\text{C} + 2\text{H}_2 \xrightarrow{t^\circ} \text{CH}_4$; $\text{C} + \text{ZnO} \xrightarrow{t^\circ} \text{Zn} + \text{CO}$
476. $\text{C} + 4\text{HNO}_3_{\text{đặc}} \xrightarrow{t^\circ} \text{CO}_2 + 4\text{NO}_2 + 2\text{H}_2\text{O}$
477. $\text{Ca(HCO}_3)_2 : \text{Ca(HCO}_3)_2 \xrightleftharpoons{t^\circ} \text{CaCO}_3 \downarrow + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
478. $\text{CaCO}_3 : \text{CaCO}_3 \xrightarrow{1000^\circ\text{C}} \text{CaO} + \text{CO}_2 \uparrow$
479. $3\text{KClO} \xrightarrow{t^\circ} \text{KClO}_3 + 2\text{KCl}$
480. $4\text{KClO}_3 \xrightarrow{t^\circ} 3\text{KClO}_4 + \text{KCl}$
481. $2\text{KClO}_3 \xrightarrow{t^\circ} 2\text{KCl} + 3\text{O}_2 \uparrow$
482. $2\text{NaHCO}_3 \xrightarrow{t^\circ} \text{Na}_2\text{CO}_3 + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
483. $2\text{NaNO}_3 \xrightarrow{t^\circ} 2\text{NaNO}_2 + \text{O}_2 \uparrow$
484. $2\text{KMnO}_4 \xrightarrow{t^\circ} \text{K}_2\text{MnO}_4 + 2\text{O}_2 \uparrow + 2\text{MnO}_2$
485. $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
486. $2\text{Fe}_2\text{O}_3 + 3\text{Si} \rightarrow 4\text{Fe} + 3\text{SiO}_2$
487. $\text{Fe}_2\text{O}_3 + 3\text{Mn} \rightarrow 2\text{Fe} + 3\text{MnO}$
488. $\text{MgSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{MgCO}_3 \downarrow + \text{Na}_2\text{SO}_4$
489. $\text{Mg(HCO}_3)_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{MgCO}_3 \downarrow + 2\text{NaHCO}_3$
490. $\text{Ca(HCO}_3)_2 \xrightarrow{t^\circ} \text{CaCO}_3 \downarrow + \text{H}_2\text{O} + \text{CO}_2 \uparrow$
491. $\text{Ca(HCO}_3)_2 + \text{Ca(OH)}_2 \rightarrow 2\text{CaCO}_3 \downarrow + \text{H}_2\text{O}$
492. $\text{FeS}_2 + 8\text{HNO}_3 \rightarrow \text{Fe(NO}_3)_3 + 2\text{H}_2\text{SO}_4 + 5\text{NO} + 2\text{H}_2\text{O}$
493. $\text{Cu(OH)}_2 + 4\text{NH}_3 \rightarrow [\text{Cu(NH}_4)_3](\text{OH})_2$
494. $\text{AgCl} + 2\text{NH}_3 \rightarrow [\text{Ag(NH}_3)_2]\text{Cl}$

495. $\text{Na}[\text{Al}(\text{OH})_4] \rightarrow \text{Na}^+ + [\text{Al}(\text{OH})_4]^-$
496. $[\text{Al}(\text{OH})_4]^- + \text{H}_3\text{O}^+ \rightarrow \text{Al}(\text{OH})_3 \downarrow + 2 \text{H}_2\text{O}$
497. $\text{NH}_4\text{Cl} + \text{Na}[\text{Al}(\text{OH})_4] \rightarrow \text{Al}(\text{OH})_3 \downarrow + \text{NH}_3 \uparrow + \text{NaCl} + \text{H}_2\text{O}$
498. $3 \text{Na}_2\text{CO}_3 + 2 \text{FeCl}_3 + 3 \text{H}_2\text{O} \rightarrow 2 \text{Fe}(\text{OH})_3 \downarrow + 3 \text{CO}_2 \uparrow + 6 \text{NaCl}$
499. $\text{Al}_2\text{S}_3 + 6 \text{H}_2\text{O} \rightarrow 2 \text{Al}(\text{OH})_3 \downarrow + 3 \text{H}_2\text{S} \uparrow$
500. $\text{Fe}_2(\text{CO}_3)_3 + 3 \text{H}_2\text{O} \rightarrow 2 \text{Fe}(\text{OH})_3 \downarrow + 3 \text{CO}_2 \uparrow$
501. $\text{NH}_4\text{Cl} + \text{NaNO}_2 \xrightarrow{t^\circ} \text{N}_2 + \text{NaCl} + 2 \text{H}_2\text{O}$
502. $\text{P} + 5 \text{HNO}_3 (\text{đặc}) \xrightarrow{t^\circ} \text{H}_3\text{PO}_4 + 5 \text{NO}_2 + \text{H}_2\text{O}$
503. $\text{NaNO}_3 (\text{rắn}) + \text{H}_2\text{SO}_4 (\text{đặc}) \xrightarrow{t^\circ} \text{HNO}_3 + \text{NaHSO}_4$
504. $\text{CO}_2 + 2 \text{NH}_3 \xrightarrow{180-200^\circ\text{C}, 200\text{atm}} (\text{NH}_2)_2\text{CO} + \text{H}_2\text{O}$
505. $5 \text{K}_2\text{SO}_3 + 2 \text{KMnO}_4 + 6 \text{KHSO}_4 \rightarrow 9 \text{K}_2\text{SO}_4 + 2 \text{MnSO}_4 + 3 \text{H}_2\text{O}$
506. $10 \text{FeCl}_2 + 6 \text{KMnO}_4 + 24 \text{H}_2\text{SO}_4 \rightarrow 5 \text{Fe}_2(\text{SO}_4)_3 + 3 \text{K}_2\text{SO}_4 + 6 \text{MnSO}_4 + 10 \text{Cl}_2 + 24 \text{H}_2\text{O}$
507. $\text{CaCO}_3 + 2 \text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$
508. $3 \text{Fe}_3\text{O}_4 + 28 \text{HNO}_3 \rightarrow 9 \text{Fe}(\text{NO}_3)_3 + \text{NO} + 14 \text{H}_2\text{O}$
509. $\text{Al} + 4 \text{HNO}_3 \rightarrow \text{Al}(\text{NO}_3)_3 + \text{NO} + 2 \text{H}_2\text{O}$
510. $\text{Al} + \text{HOH} + \text{NaOH} \rightarrow \text{NaAlO}_2 + 3/2 \text{H}_2$
511. $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$
512. $\text{Fe}_3\text{O}_4 + 4 \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{Fe}_2(\text{SO}_4)_3 + 4 \text{H}_2\text{O}$
513. $\text{CaCO}_3 \xrightarrow{t^\circ} \text{CaO} + \text{CO}_2$
 $\text{Fe}_3\text{O}_4 + 4 \text{CO} \xrightarrow{t^\circ} 3 \text{Fe} + 4 \text{CO}_2$
514. $\text{Ca}_3(\text{PO}_4)_2 + 5 \text{C} + 3 \text{SiO}_2 \xrightarrow{1200^\circ\text{C}} 3 \text{CaSiO}_3 + 2 \text{P} + 5 \text{CO}$
515. $\text{Cl}_2 + 2 \text{NaBr} \longrightarrow 2 \text{NaCl} + \text{Br}_2$
516. $\text{Cl}_2 + 2 \text{NaI} \longrightarrow 2 \text{NaCl} + \text{I}_2$
517. $\text{Cl}_2 + 2 \text{H}_2\text{O} + \text{SO}_2 \longrightarrow 2 \text{HCl} + \text{H}_2\text{SO}_4$
518. $\text{Cl}_2 + 2 \text{FeCl}_2 \longrightarrow 2 \text{FeCl}_3$
519. $\text{MnO}_2 + 4 \text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2 \text{H}_2\text{O}$
520. $2 \text{KMnO}_4 + 16 \text{HCl} \rightarrow 2 \text{MnCl}_2 + 5 \text{Cl}_2 + 2 \text{KCl} + 8 \text{H}_2\text{O}$
521. $\text{KClO}_3 + 6 \text{HCl} \rightarrow \text{KCl} + 3 \text{H}_2\text{O} + 3 \text{Cl}_2$
522. $2 \text{NaCl} + 2 \text{H}_2\text{O} \longrightarrow 2 \text{NaOH} + \text{H}_2 + \text{Cl}_2$
523. dpnc
524. $2 \text{NaCl} \longrightarrow \text{Na} + \text{Cl}_2$
525. $2 \text{KMnO}_4 + 16 \text{HCl} \longrightarrow 2 \text{KCl} + 5 \text{Cl}_2 + 2 \text{MnCl}_2 + 8 \text{H}_2\text{O}$
526. $2 \text{NaCl}_{(\text{R})} + \text{H}_2\text{SO}_4_{\text{đc}} \longrightarrow \text{Na}_2\text{SO}_4 + 2 \text{HCl}$
527. $\text{HCl} + \text{AgNO}_3 \longrightarrow \text{AgCl} + \text{HNO}_3$
528. $\text{NaCl} + \text{AgNO}_3 \longrightarrow \text{AgCl} + \text{NaNO}_3$
529. $3 \text{Cl}_2 + 6 \text{KOH} \longrightarrow 5 \text{KCl} + \text{KClO}_3 + 3 \text{H}_2\text{O}$
530. $\text{F}_2 + \text{H}_2\text{S} \rightarrow 2 \text{HF} + \text{S}$
531. $\text{F}_2 + \text{H}_2\text{O} \rightarrow \text{HF} + \text{O}_2$
532. $\text{Cl}_2 + \text{H}_2\text{S} \rightarrow 2 \text{HCl} + \text{S}$
533. $3 \text{FeCl}_2 + 3 \text{Cl}_2 \rightarrow 2 \text{FeCl}_3$
534. $\text{Cl}_2 + 2 \text{NaBr} \rightarrow 2 \text{NaCl} + \text{Br}_2$
535. $\text{Cl}_2 + 2 \text{NaI} \rightarrow 2 \text{NaCl} + \text{I}_2$
536. $\text{Br}_2 + \text{H}_2 \rightarrow 2 \text{HBr}$
537. $\text{Br}_2 + 2 \text{NaI} \rightarrow 2 \text{NaBr} + \text{I}_2$
538. $\text{I}_2 + \text{H}_2\text{S} \rightarrow 2 \text{HI} + \text{S}$
539. $3 \text{Cl}_2 + \text{S} + 4 \text{H}_2\text{O} \rightarrow 6 \text{HCl} + \text{H}_2\text{SO}_4$
540. $\text{Cl}_2 + \text{SO}_2 + 2 \text{H}_2\text{O} \rightarrow 2 \text{HCl} + \text{H}_2\text{SO}_4$
541. $4 \text{Cl}_2 + \text{H}_2\text{S} + 4 \text{H}_2\text{O} \rightarrow 8 \text{HCl} + \text{H}_2\text{SO}_4$

542. $3\text{Br}_2 + \text{S} + 4\text{H}_2\text{O} \rightarrow 6\text{HBr} + \text{H}_2\text{SO}_4$
543. $\text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{HBr} + \text{H}_2\text{SO}_4$ (phản ứng nhận biết khí SO_2).
544. $4\text{Br}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O} \rightarrow 8\text{HBr} + \text{H}_2\text{SO}_4$
545. $5\text{Cl}_2 + 6\text{H}_2\text{O} + \text{I}_2 \rightarrow 10\text{HCl} + 2\text{HIO}_3$
546. $5\text{Br}_2 + 6\text{H}_2\text{O} + \text{I}_2 \rightarrow 10\text{HBr} + 2\text{HIO}_3$
547. $3\text{Cl}_2 + 6\text{NaOH} \xrightarrow{70^\circ} 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$
548. $\text{Cl}_2 + \text{Ca}(\text{OH})_2 \xrightarrow{30^\circ} \text{CaOCl}_2 + \text{H}_2\text{O}$
549. $3\text{Br}_2 + 6\text{NaOH} \rightarrow 5\text{NaBr} + \text{NaBrO}_3 + 3\text{H}_2\text{O}$
550. $\text{MnO}_2 + 4\text{HCl} \xrightarrow{t^\circ} \text{MnCl}_2 + \text{Cl}_2 \uparrow + 2\text{H}_2\text{O}$
551. $\text{MnO}_2 + 4\text{NaCl} + 4\text{H}_2\text{SO}_4 \xrightarrow{t^\circ} \text{MnCl}_2 + 4\text{NaHSO}_4 + \text{Cl}_2 \uparrow + 2\text{H}_2\text{O}$
552. $2\text{KMnO}_4 + 16\text{HCl} \xrightarrow{t^\circ} 2\text{MnCl}_2 + \text{Cl}_2 \uparrow + 2\text{KCl} + 8\text{H}_2\text{O}$
553. $\text{K}_2\text{Cr}_2\text{O}_7 + 14\text{HCl} \xrightarrow{t^\circ} 2\text{KCl} + 2\text{CrCl}_3 + 3\text{Cl}_2 \uparrow + 7\text{H}_2\text{O}$
554. $\text{CaOCl}_2 + 2\text{HCl} \xrightarrow{t^\circ} \text{CaCl}_2 + \text{H}_2\text{O} + \text{Cl}_2 \uparrow$
555. $2\text{NaClO} + 2\text{HCl} \xrightarrow{t^\circ} 2\text{NaCl} + \text{Cl}_2 \uparrow + \text{H}_2\text{O}$
556. $2\text{NaI} + \text{MnO}_2 + 2\text{H}_2\text{SO}_4 \xrightarrow{t^\circ} \text{MnSO}_4 + \text{I}_2 + \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
557. $2\text{NaBr} + \text{MnO}_2 + 2\text{H}_2\text{SO}_4 \xrightarrow{t^\circ} \text{MnSO}_4 + \text{Br}_2 + \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
558. $2\text{HCl} + \text{Cu}(\text{OH})_2 \rightarrow \text{CuCl}_2 + 2\text{H}_2\text{O}$
559. $\text{HBr} + \text{NaOH} \rightarrow \text{NaBr} + \text{H}_2\text{O}$
560. $8\text{HCl} + \text{Fe}_3\text{O}_4 \rightarrow 2\text{FeCl}_3 + \text{FeCl}_2 + 4\text{H}_2\text{O}$
561. $8\text{HBr} + \text{Fe}_3\text{O}_4 \rightarrow 2\text{FeBr}_3 + \text{FeBr}_2 + 4\text{H}_2\text{O}$
562. $8\text{HI} + \text{Fe}_3\text{O}_4 \rightarrow 2\text{FeI}_3 + \text{FeI}_2 + 4\text{H}_2\text{O}$
563. $\text{MnO}_2 + 4\text{HCl} \xrightarrow{t^\circ} \text{MnCl}_2 + \text{Cl}_2 \uparrow + 2\text{H}_2\text{O}$
564. $\text{MnO}_2 + 4\text{NaCl} + 4\text{H}_2\text{SO}_4 \xrightarrow{t^\circ} \text{MnCl}_2 + 4\text{NaHSO}_4 + \text{Cl}_2 \uparrow + 2\text{H}_2\text{O}$
565. $2\text{KMnO}_4 + 16\text{HCl} \xrightarrow{t^\circ} 2\text{MnCl}_2 + \text{Cl}_2 \uparrow + 2\text{KCl} + 8\text{H}_2\text{O}$
566. $\text{K}_2\text{Cr}_2\text{O}_7 + 14\text{HCl} \xrightarrow{t^\circ} 2\text{KCl} + 2\text{CrCl}_3 + 3\text{Cl}_2 \uparrow + 7\text{H}_2\text{O}$
567. $\text{CaOCl}_2 + 2\text{HCl} \xrightarrow{t^\circ} \text{CaCl}_2 + \text{H}_2\text{O} + \text{Cl}_2 \uparrow$
568. $2\text{NaClO} + 2\text{HCl} \xrightarrow{t^\circ} 2\text{NaCl} + \text{Cl}_2 \uparrow + \text{H}_2\text{O}$
569. $\text{PbO}_2 + 4\text{HCl} \xrightarrow{t^\circ} \text{PbCl}_2 + \text{Cl}_2 \uparrow + 2\text{H}_2\text{O}$
570. $2\text{HBr}(\text{k}) + \text{H}_2\text{SO}_4(\text{®}) \rightarrow \text{Br}_2 + \text{SO}_2 \uparrow + 2\text{H}_2\text{O}$
571. $8\text{HI}(\text{k}) + \text{H}_2\text{SO}_4(\text{®}) \rightarrow 4\text{I}_2 + \text{H}_2\text{S} \uparrow + 4\text{H}_2\text{O}$
572. $4\text{HBr} + \text{O}_2 \rightarrow \text{Br}_2 + 2\text{H}_2\text{O}$
573. $4\text{HI} + \text{O}_2 \rightarrow \text{I}_2 + 2\text{H}_2\text{O}$
574. $\text{MnO}_2 + 4\text{HBr} \rightarrow \text{MnBr}_2 + \text{Br}_2 + 2\text{H}_2\text{O}$
575. $\text{MnO}_2 + 4\text{HI} \rightarrow \text{MnI}_2 + \text{I}_2 \uparrow + 2\text{H}_2\text{O}$
576. $4\text{HF} + \text{SiO}_2 \rightarrow \text{SiF}_4 \uparrow + 2\text{H}_2\text{O}$
577. $\text{CaF}_2(\text{r/m}) + \text{H}_2\text{SO}_4(\text{®}) \rightarrow \text{CaSO}_4 + 2\text{HF} \uparrow$
578. $\text{NaCl}(\text{r/m}) + \text{H}_2\text{SO}_4(\text{®}) \xrightarrow{250^\circ} \text{NaHSO}_4 + \text{HCl} \uparrow$
579. $2\text{NaCl}(\text{r/m}) + \text{H}_2\text{SO}_4(\text{®}) \xrightarrow{>250^\circ} \text{Na}_2\text{SO}_4 + 2\text{HCl} \uparrow$
580. $2\text{NaBr}(\text{k}) + 2\text{H}_2\text{SO}_4(\text{®}) \rightarrow \text{Br}_2 \uparrow + \text{SO}_2 \uparrow + 2\text{H}_2\text{O} + \text{Na}_2\text{SO}_4$
581. $8\text{NaI}(\text{k}) + 5\text{H}_2\text{SO}_4(\text{®}) \rightarrow 4\text{I}_2 \uparrow + \text{H}_2\text{S} \uparrow + 4\text{H}_2\text{O} + 4\text{Na}_2\text{SO}_4$
582. $\text{PBr}_3 + 3\text{H}_2\text{O} \rightarrow 3\text{HBr} + \text{H}_3\text{PO}_3$
583. $\text{PI}_3 + 3\text{H}_2\text{O} \rightarrow 3\text{HI} + \text{H}_3\text{PO}_3$

584. $\text{H}_2\text{S} + \text{I}_2 \rightarrow \text{S} \downarrow + 2\text{HI}$
585. $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$
586. $\text{NaBr} + \text{AgNO}_3 \rightarrow \text{AgBr} \downarrow + \text{NaNO}_3$
587. $\text{NaI} + \text{AgNO}_3 \rightarrow \text{AgI} \downarrow + \text{NaNO}_3$
588. $2\text{SO}_2 + \text{O}_2 \xrightarrow{\text{V}_2\text{O}_5, 300^\circ\text{C}} 2\text{SO}_3$
589. $\text{CH}_4 + 2\text{O}_2 \xrightarrow{t^\circ} \text{CO}_2 + 2\text{H}_2\text{O}$
590. $\text{O}_3 + 2\text{KI} + \text{H}_2\text{O} \longrightarrow \text{I}_2 + 2\text{KOH} + \text{O}_2$ (oxi không có)
591. $2\text{Ag} + \text{O}_3 \longrightarrow \text{Ag}_2\text{O} + \text{O}_2$ (oxi không có phản ứng)
592. $2\text{H}_2\text{S} + 3\text{O}_2 \xrightarrow{t^\circ} 2\text{H}_2\text{O} + 2\text{SO}_2$ (dư oxi, đốt cháy)
593. $2\text{H}_2\text{S} + \text{O}_2 \xrightarrow{t^\circ \text{ áp suất}} 2\text{H}_2\text{O} + 2\text{S} \downarrow$
594. $\text{H}_2\text{S} + 4\text{Cl}_2 + 4\text{H}_2\text{O} \longrightarrow 8\text{HCl} + \text{H}_2\text{SO}_4$
595. $\text{H}_2\text{S} + \text{Cl}_2 \longrightarrow 2\text{HCl} + \text{S}$ (khí clo gặp khí H_2S)
596. $\text{H}_2\text{S} + \text{NaOH} \xrightarrow{\text{t}^\circ} \text{NaHS} + \text{H}_2\text{O}$
597. $\text{H}_2\text{S} + 2\text{NaOH} \xrightarrow{1:2} \text{Na}_2\text{S} + 2\text{H}_2\text{O}$
598. $\text{SO}_2 + 2\text{NaOH} \xrightarrow{1:2} \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$
599. $\text{SO}_2 + \text{NaOH} \xrightarrow{\text{t}^\circ} \text{NaHSO}_3$
600. $\text{H}_2\text{SO}_4 \longrightarrow 2\text{H}^+ + \text{SO}_4^{2-}$ là quì tím hoá màu đỏ.
601. $\text{H}_2\text{SO}_4 + \text{Fe} \longrightarrow \text{FeSO}_4 + \text{H}_2 \uparrow$
602. $\text{H}_2\text{SO}_4 + \text{NaOH} \longrightarrow \text{NaHSO}_4 + \text{H}_2\text{O}$
603. $\text{H}_2\text{SO}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
604. $\text{H}_2\text{SO}_4 + \text{CuO} \longrightarrow \text{CuSO}_4 + \text{H}_2\text{O}$
605. $\text{H}_2\text{SO}_4 + \text{BaCl}_2 \longrightarrow \text{BaSO}_4 \downarrow + 2\text{HCl}$
606. $\text{H}_2\text{SO}_4 + \text{Na}_2\text{SO}_3 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{SO}_2 \uparrow$
607. $\text{H}_2\text{SO}_4 + \text{CaCO}_3 \longrightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2 \uparrow$
608. $2\text{Fe} + 6\text{H}_2\text{SO}_4 \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + 3\text{SO}_2 + 6\text{H}_2\text{O}$
609. $\text{Cu} + 2\text{H}_2\text{SO}_4 \xrightarrow{t^\circ} \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
610. $2\text{H}_2\text{SO}_{4(d)} + \text{C} \xrightarrow{t^\circ} \text{CO}_2 + 2\text{SO}_2 + 2\text{H}_2\text{O}$
611. $2\text{H}_2\text{SO}_{4(d)} + \text{S} \xrightarrow{t^\circ} 3\text{SO}_2 + 2\text{H}_2\text{O}$
612. $\text{FeO} + \text{H}_2\text{SO}_{4(d)} \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 4\text{H}_2\text{O}$
613. $2\text{HBr} + \text{H}_2\text{SO}_{4(d)} \xrightarrow{t^\circ} \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
614. $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{SO}_{4(d)} \longrightarrow 12\text{C} + \text{H}_2\text{SO}_4 + 11\text{H}_2\text{O}$
615. $\text{S} + \text{O}_2 \xrightarrow{t^\circ} \text{SO}_2$
616. $\text{Na}_2\text{SO}_3 + \text{H}_2\text{SO}_{4(d)} \xrightarrow{t^\circ} \text{Na}_2\text{SO}_4 + \text{H}_2\text{O} + \text{SO}_2 \uparrow$
617. $\text{Cu} + 2\text{H}_2\text{SO}_{4(d)} \xrightarrow{t^\circ} \text{CuSO}_4 + 2\text{H}_2\text{O} + \text{SO}_2 \uparrow$
618. $4\text{FeS}_2 + 11\text{O}_2 \xrightarrow{t^\circ} 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$
619. $2\text{SO}_2 + \text{O}_2 \longrightarrow 2\text{SO}_3$ (xúc tác V_2O_5 , t°)
620. $4\text{FeS}_2 + 11\text{O}_2 \xrightarrow{t^\circ} 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$
621. $\text{SO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4$
622. $2\text{SO}_2 + \text{O}_2 \xrightarrow[\text{25}]{\text{V}_2\text{O}_5, t^\circ} 2\text{SO}_3$
623. $2\text{NaCl} + \text{H}_2\text{O} \xrightarrow{\text{điện phân}} \text{H}_2 + \text{Cl}_2 + 2\text{NaOH}$
624. $2\text{NaHCO}_3 \xrightarrow{t^\circ} \text{Na}_2\text{CO}_3 + \text{Na}_2\text{CO}_3 + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
625. $\text{NaHCO}_3 + \text{HCl} = \text{NaCl} + \text{CO}_2 \uparrow + \text{H}_2\text{O}$
626. $\text{NaHCO}_3 + \text{NaOH} = \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$

627. $\text{Na}_2\text{CO}_3 + 2\text{HCl} = 2\text{NaCl} + \text{CO}_2\uparrow + \text{H}_2\text{O}$
628. $4\text{M} + 10\text{HNO}_3 = 4\text{M}(\text{NO}_3)_2 + 3\text{H}_2\text{O} + \text{NH}_4\text{NO}_3$
629. $\text{Ca}(\text{OH})_2 + \text{Na}_2\text{CO}_3 = \text{CaCO}_3\downarrow + 2\text{NaOH}$
630. $\text{CaCO}_3 + 2\text{HCl} = \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2\uparrow$
631. $\text{CaCO}_3 + 2\text{CH}_3\text{COOH} = \text{Ca}(\text{CH}_3\text{COO})_2 + \text{H}_2\text{O} + \text{CO}_2\uparrow$
632. $\text{Ca}(\text{HCO}_3)_2 \xrightarrow{p} \text{CaCO}_3\downarrow + \text{H}_2\text{O} + \text{CO}_2\uparrow$
633. $\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 = 2\text{CaCO}_3\downarrow + 2\text{H}_2\text{O}$
634. $\text{CaSO}_4 + \text{Na}_2\text{CO}_3 = \text{CaSO}_3\downarrow + \text{Na}_2\text{SO}_4$
635. $\text{Ca}(\text{HCO}_3)_2 + \text{Na}_2\text{CO}_3 = \text{CaCO}_3\downarrow + 2\text{NaHCO}_3$
636. $\text{Al}(\text{OH})_3 + 3\text{HCl} = \text{AlCl}_3 + 3\text{H}_2\text{O}$
637. $\text{HAlO}_2 \cdot \text{H}_2\text{O} + \text{OH}^- = \text{AlO}_2^- + 2\text{H}_2\text{O}$
638. $\text{Al}(\text{OH})_3 + \text{NaOH} = \text{NaAlO}_2 + 2\text{H}_2\text{O}$
639. $\text{AlCl}_3 + 3\text{NH}_4\text{OH} = \text{Al}(\text{OH})_3\downarrow + 3\text{NH}_4\text{Cl}$
640. $3\text{CaO} + \text{P}_2\text{O}_5 = \text{Ca}_3(\text{PO}_4)_2$
641. $3\text{Fe} + 4\text{H}_2\text{O} \xrightarrow{<500^\circ\text{C}} \text{Fe}_3\text{O}_4 + 4\text{H}_2$
642. $\text{Fe} + \text{H}_2\text{O} \xrightarrow{>500^\circ\text{C}} \text{FeO} + \text{H}_2$
643. $3\text{FeO} + 10\text{HNO}_3 \text{ đặc} \xrightarrow{p} 3\text{Fe}(\text{NO}_3)_3 + \text{NO} + 5\text{H}_2\text{O}$
644. $2\text{FeO} + 4\text{H}_2\text{SO}_4 \text{ đặc} \xrightarrow{p} \text{Fe}_2(\text{SO}_4)_3 + \text{SO}_2 + 4\text{H}_2\text{O}$
645. $\text{FeO} + \text{H}_2\text{SO}_4 \text{ loãng} \longrightarrow \text{FeSO}_4 + \text{H}_2\text{O}$
646. $\text{FeO} + 2\text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2\text{O}$
647. $\text{FeO} + \text{CO} \xrightarrow{p} \text{Fe} + \text{CO}_2$
648. $\text{Fe}(\text{OH})_2 + 2\text{HCl} \longrightarrow \text{FeCl}_2 + 2\text{H}_2\text{O}$
649. $\text{Fe}(\text{OH})_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{FeSO}_4 + 2\text{H}_2\text{O}$
650. $4\text{Fe}(\text{OH})_2 + \text{O}_2 + 2\text{H}_2\text{O} \longrightarrow 4\text{Fe}(\text{OH})_3$
651. $\text{FeCl}_2 + 2\text{NaOH} \longrightarrow \text{Fe}(\text{OH})_2 + 2\text{NaCl}$
652. $2\text{FeCl}_2 + \text{Cl}_2 \longrightarrow 2\text{FeCl}_3$
653. $10\text{FeSO}_4 + 2\text{KMnO}_4 + 8\text{H}_2\text{SO}_4 \longrightarrow 5\text{Fe}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 8\text{H}_2\text{O}$
654. $3\text{Fe}_2\text{O}_3 + \text{CO} \xrightarrow{p} 2\text{Fe}_3\text{O}_4 + \text{CO}_2$
655. $\text{Fe}_2\text{O}_3 + \text{CO} \xrightarrow{p} 2\text{FeO} + \text{CO}_2$
656. $\text{Fe}_2\text{O}_3 + 3\text{CO} \xrightarrow{p} 2\text{Fe} + 3\text{CO}_2$
657. $\text{Fe}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \text{ loãng} \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$
658. $\text{Fe}_2\text{O}_3 + 6\text{HCl} \longrightarrow 2\text{FeCl}_3 + 3\text{H}_2\text{O}$
659. $\text{Fe}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$
660. $\text{FeCl}_3 + 3\text{NaOH} \longrightarrow \text{Fe}(\text{OH})_3 + 3\text{NaCl}$
661. $2\text{FeCl}_3 + \text{Fe} \longrightarrow 3\text{FeCl}_2$
662. $2\text{FeCl}_3 + \text{Cu} \longrightarrow 2\text{FeCl}_2 + \text{CuCl}_2$
663. $2\text{FeCl}_3 + 2\text{KI} \longrightarrow 2\text{FeCl}_2 + 2\text{KCl} + \text{I}_2$
664. $2\text{Fe}(\text{OH})_3 \xrightarrow{p} \text{Fe}_2\text{O}_3 + 3\text{H}_2\text{O}$
665. $2\text{Fe}(\text{OH})_3 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 6\text{H}_2\text{O}$
666. $\text{Fe}(\text{OH})_3 + 3\text{HCl} \longrightarrow \text{FeCl}_3 + 3\text{H}_2\text{O}$
667. $2\text{FeS}_2 + 14\text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 15\text{SO}_2 + 14\text{H}_2\text{O}$
668. $4\text{FeS}_2 + 11\text{O}_2 \xrightarrow{p} 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$
669. $4\text{Cr} + 3\text{O}_2 \xrightarrow{p} 2\text{Cr}_2\text{O}_3$
670. $2\text{Cr} + 3\text{Cl}_2 \xrightarrow{p} 2\text{CrCl}_3$
671. $2\text{Cr} + 3\text{S} \xrightarrow{p} \text{Cr}_2\text{S}_3$

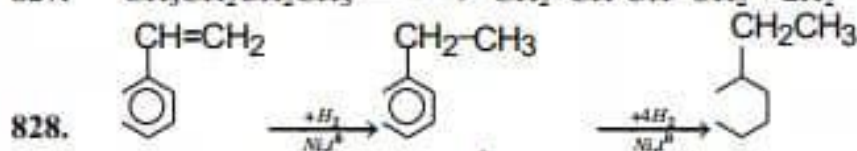
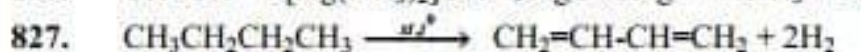
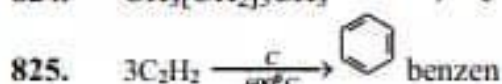
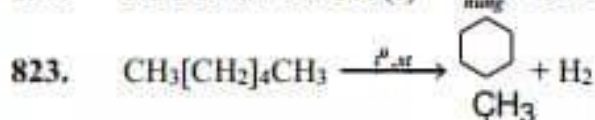
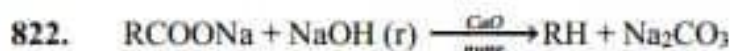
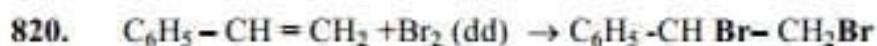
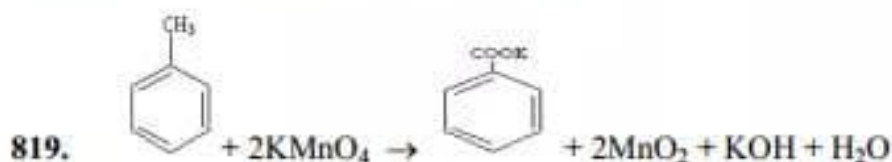
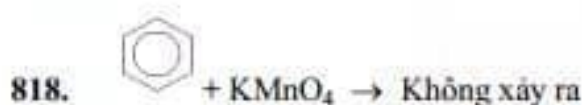
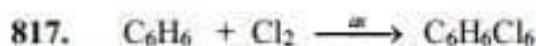
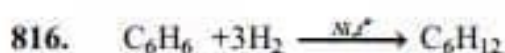
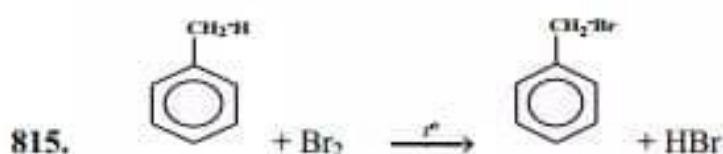
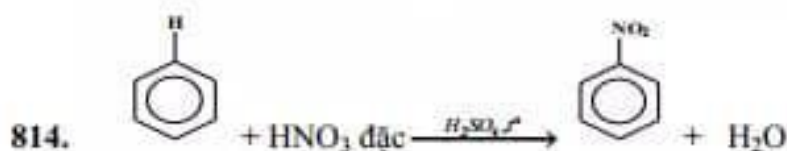
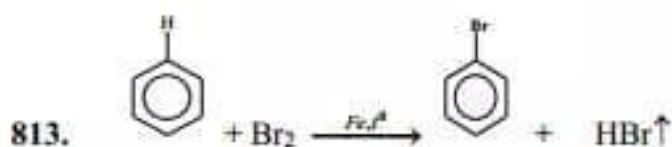
672. $\text{Cr} + 2\text{HCl} \longrightarrow \text{CrCl}_2 + \text{H}_2$.
673. $\text{Cr} + \text{H}_2\text{SO}_4 \longrightarrow \text{CrSO}_4 + \text{H}_2$.
674. $2\text{Cr} + 3\text{SnCl}_2 \longrightarrow 2\text{CrCl}_3 + 3\text{Sn}$.
675. $4\text{Cr}(\text{OH})_2 + \text{O}_2 + 2\text{H}_2\text{O} \xrightarrow{t^\circ} 4\text{Cr}(\text{OH})_3$.
676. $\text{Cr}(\text{OH})_2 + 2\text{HCl} \longrightarrow \text{CrCl}_2 + 2\text{H}_2\text{O}$.
677. $\text{Cr}(\text{OH})_3 + \text{NaOH} \longrightarrow \text{Na}[\text{Cr}(\text{OH})_4] \text{ (hay } \text{NaCrO}_2\text{)}$.
678. $\text{Cr}(\text{OH})_3 + 3\text{HCl} \longrightarrow \text{CrCl}_3 + 3\text{H}_2\text{O}$.
679. $2\text{Cr}(\text{OH})_3 \xrightarrow{t^\circ} \text{Cr}_2\text{O}_3 + 3\text{H}_2\text{O}$.
680. $2\text{CrO} + \text{O}_2 \xrightarrow{>100^\circ\text{C}} 2\text{Cr}_2\text{O}_3$.
681. $\text{CrO} + 2\text{HCl} \longrightarrow \text{CrCl}_2 + \text{H}_2\text{O}$.
682. $\text{Cr}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$.
683. $2\text{Cr}_2\text{O}_3 + 8\text{NaOH} + 3\text{O}_2 \longrightarrow 4\text{Na}_2\text{CrO}_4 + 4\text{H}_2\text{O}$.
684. $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{t^\circ} 2\text{Cr} + \text{Al}_2\text{O}_3$.
685. $\text{CrO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{CrO}_4$.
686. $2\text{CrO}_3 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{Cr}_2\text{O}_7$.
687. $4\text{CrO}_3 \xrightarrow{420^\circ\text{C}} 2\text{Cr}_2\text{O}_3 + 3\text{O}_2$.
688. $2\text{CrO}_3 + 2\text{NH}_3 \longrightarrow \text{Cr}_2\text{O}_3 + \text{N}_2 + 3\text{H}_2\text{O}$.
689. $4\text{CrCl}_2 + \text{O}_2 + 4\text{HCl} \longrightarrow 4\text{CrCl}_3 + 2\text{H}_2\text{O}$.
690. $\text{CrCl}_2 + 2\text{NaOH} \longrightarrow \text{Cr}(\text{OH})_2 + 2\text{NaCl}$.
691. $2\text{CrCl}_2 + \text{Cl}_2 \longrightarrow 2\text{CrCl}_3$.
692. $2\text{CrCl}_3 + \text{Zn} \longrightarrow \text{ZnCl}_2 + 2\text{CrCl}_2$.
693. $\text{CrCl}_3 + 3\text{NaOH} \longrightarrow \text{Cr}(\text{OH})_3 + 3\text{NaCl}$.
694. $2\text{CrCl}_3 + 3\text{Cl}_2 + 16\text{NaOH} \longrightarrow 2\text{Na}_2\text{CrO}_4 + 12\text{NaCl} + 8\text{H}_2\text{O}$.
695. $2\text{NaCrO}_2 + 3\text{Br}_2 + 8\text{NaOH} \longrightarrow 2\text{Na}_2\text{CrO}_4 + 6\text{NaBr} + 4\text{H}_2\text{O}$.
696. $2\text{Na}_2\text{Cr}_2\text{O}_7 + 3\text{C} \longrightarrow 2\text{Na}_2\text{CO}_3 + \text{CO}_2 + 2\text{Cr}_2\text{O}_3$.
697. $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{S} \longrightarrow \text{Na}_2\text{SO}_4 + \text{Cr}_2\text{O}_3$.
698. $\text{Na}_2\text{Cr}_2\text{O}_7 + 14\text{HCl} \longrightarrow 2\text{CrCl}_3 + 2\text{NaCl} + 3\text{Cl}_2 + 7\text{H}_2\text{O}$.
699. $\text{K}_2\text{Cr}_2\text{O}_7 + 3\text{H}_2\text{S} + 4\text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + 3\text{S} + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$.
700. $\text{K}_2\text{Cr}_2\text{O}_7 + 3\text{K}_2\text{SO}_3 + 4\text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + 4\text{K}_2\text{SO}_4 + 4\text{H}_2\text{O}$.
701. $\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{KI} + 7\text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + 4\text{K}_2\text{SO}_4 + 3\text{I}_2 + 7\text{H}_2\text{O}$.
702. $\text{K}_2\text{Cr}_2\text{O}_7 + 6\text{FeSO}_4 + 7\text{H}_2\text{SO}_4 \longrightarrow 3\text{Fe}_2(\text{SO}_4)_3 + \text{Cr}_2(\text{SO}_4)_3 + \text{K}_2\text{SO}_4 + 7\text{H}_2\text{O}$.
703. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{t^\circ} \text{Cr}_2\text{O}_3 + \text{N}_2 + 4\text{H}_2\text{O}$.
704. $2\text{Na}_2\text{Cr}_2\text{O}_7 \xrightarrow{t^\circ} 2\text{Na}_2\text{O} + 2\text{Cr}_2\text{O}_3 + 3\text{O}_2$.
705. $2\text{Na}_2\text{CrO}_4 + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$.
706. $\text{Cu} + \text{Cl}_2 \xrightarrow{t^\circ} \text{CuCl}_2$.
707. $2\text{Cu} + \text{O}_2 \xrightarrow{t^\circ} 2\text{CuO}$.
708. $\text{Cu} + \text{S} \xrightarrow{t^\circ} \text{CuS}$.
709. $\text{Cu} + 2\text{H}_2\text{SO}_4 \text{ đ\acute{a}c} \longrightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$.
710. $\text{Cu} + 4\text{HNO}_3 \text{ đ\acute{a}c} \longrightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$.
711. $3\text{Cu} + 8\text{HNO}_3 \text{ lo\`ang} \longrightarrow 3\text{Cu}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$.
712. $\text{Cu} + 2\text{AgNO}_3 \longrightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$.
713. $\text{Cu} + 2\text{FeCl}_3 \longrightarrow \text{CuCl}_2 + 2\text{FeCl}_2$.
714. $3\text{Cu} + 8\text{NaNO}_3 + 4\text{H}_2\text{SO}_4 \longrightarrow 3\text{Cu}(\text{NO}_3)_2 + 4\text{Na}_2\text{SO}_4 + 2\text{NO} + 4\text{H}_2\text{O}$.
715. $2\text{Cu} + 4\text{HCl} + \text{O}_2 \longrightarrow 2\text{CuCl}_2 + 2\text{H}_2\text{O}$.
716. 89. $\text{CuO} + \text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + \text{H}_2\text{O}$.

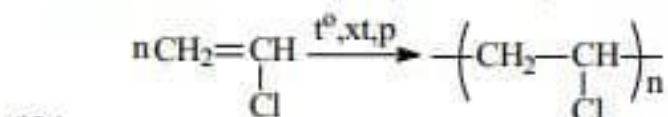
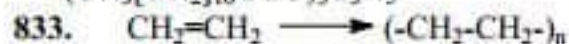
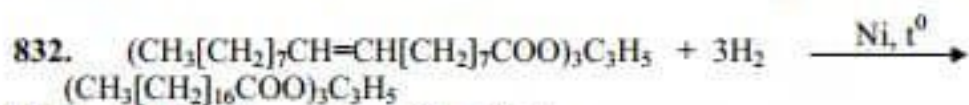
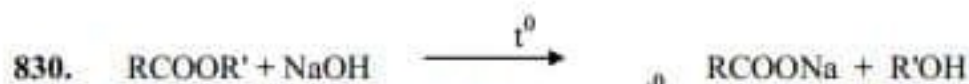
717. $\text{CuO} + 2\text{HCl} \longrightarrow \text{CuCl}_2 + \text{H}_2\text{O}$.
718. $\text{CuO} + \text{H}_2 \xrightarrow{t^\circ} \text{Cu} + \text{H}_2\text{O}$.
719. $\text{CuO} + \text{CO} \xrightarrow{t^\circ} \text{Cu} + \text{CO}_2$.
720. $3\text{CuO} + 2\text{NH}_3 \xrightarrow{t^\circ} \text{N}_2 + 3\text{Cu} + 3\text{H}_2\text{O}$.
721. $\text{CuO} + \text{Cu} \xrightarrow{t^\circ} \text{Cu}_2\text{O}$.
722. $\text{Cu}_2\text{O} + \text{H}_2\text{SO}_4 \text{ loãng} \longrightarrow \text{CuSO}_4 + \text{Cu} + \text{H}_2\text{O}$.
723. $\text{Cu}(\text{OH})_2 + 2\text{HCl} \longrightarrow \text{CuCl}_2 + 2\text{H}_2\text{O}$.
724. $\text{Cu}(\text{OH})_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{CuSO}_4 + 2\text{H}_2\text{O}$.
725. $\text{Cu}(\text{OH})_2 \xrightarrow{t^\circ} \text{CuO} + \text{H}_2\text{O}$.
726. $\text{Cu}(\text{OH})_2 + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + 2\text{OH}^-$.
727. $2\text{Cu}(\text{NO}_3)_2 \xrightarrow{t^\circ} 2\text{CuO} + 2\text{NO}_2 + 3\text{O}_2$.
728. $\xrightarrow{\text{điện phân dung dịch}} \text{Cu} + \text{Cl}_2$.
729. $2\text{Cu}(\text{NO}_3)_2 + 2\text{H}_2\text{O} \xrightarrow{\text{điện phân dung dịch}} 2\text{Cu} + 4\text{HNO}_3 + \text{O}_2$.
730. $2\text{CuSO}_4 + 2\text{H}_2\text{O} \xrightarrow{\text{điện phân dung dịch}} 2\text{Cu} + 2\text{H}_2\text{SO}_4 + \text{O}_2$.
731. $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2 \xrightarrow{t^\circ} 2\text{CuO} + \text{CO}_2 + \text{H}_2\text{O}$.
732. $\text{CuS} + 2\text{AgNO}_3 \longrightarrow 2\text{AgS} + \text{Cu}(\text{NO}_3)_2$.
733. $\text{CuS} + 4\text{H}_2\text{SO}_4 \text{ đặc} \longrightarrow \text{CuSO}_4 + 4\text{SO}_2 + 4\text{H}_2\text{O}$.
734. $2\text{Ni} + \text{O}_2 \xrightarrow{500^\circ\text{C}} 2\text{NiO}$.
735. $\text{Ni} + \text{Cl}_2 \xrightarrow{t^\circ} \text{NiCl}_2$.
736. $\text{Zn} + \text{O}_2 \xrightarrow{t^\circ} 2\text{ZnO}$.
737. $\text{Zn} + \text{S} \xrightarrow{t^\circ} \text{ZnS}$.
738. $\text{Zn} + \text{Cl}_2 \xrightarrow{t^\circ} \text{ZnCl}_2$.
739. $2\text{Pb} + \text{O}_2 \xrightarrow{t^\circ} 2\text{PbO}$.
740. $\text{Pb} + \text{S} \xrightarrow{t^\circ} \text{PbS}$.
741. $3\text{Pb} + 8\text{HNO}_3 \text{ loãng} \longrightarrow 3\text{Pb}(\text{NO}_3)_2 + 2\text{NO} + 4\text{H}_2\text{O}$.
742. $\text{Sn} + 2\text{HCl} \longrightarrow \text{SnCl}_2 + \text{H}_2$.
743. $\text{Sn} + \text{O}_2 \xrightarrow{t^\circ} \text{SnO}_2$.
744. $5\text{Sn}^{2+} + 2\text{MnO}_4^- + 16\text{H}^+ \rightarrow 5\text{Sn}^{4+} + 2\text{Mn}^{2+} + 8\text{H}_2\text{O}$.
745. $\text{Ag} + 2\text{HNO}_3 \text{ (đặc)} \longrightarrow \text{AgNO}_3 + \text{NO}_2 + \text{H}_2\text{O}$.
746. $2\text{Ag} + 2\text{H}_2\text{S} + \text{O}_2 \longrightarrow 2\text{Ag}_2\text{S} + 2\text{H}_2\text{O}$.
747. $2\text{Ag} + \text{O}_3 \longrightarrow \text{Ag}_2\text{O} + \text{O}_2$.
748. $\text{Ag}_2\text{O} + \text{H}_2\text{O}_2 \longrightarrow 2\text{Ag} + \text{H}_2\text{O} + \text{O}_2$.
749. $2\text{AgNO}_3 \xrightarrow{t^\circ} 2\text{Ag} + 2\text{NO}_2 + \text{O}_2$.
750. $4\text{AgNO}_3 + 2\text{H}_2\text{O} \xrightarrow{\text{điện phân dung dịch}} 4\text{Ag} + 4\text{HNO}_3 + \text{O}_2$.
751. $\text{Au} + \text{HNO}_3 + 3\text{HCl} \longrightarrow \text{AuCl}_3 + 2\text{H}_2\text{O} + \text{NO}$.
752. $\text{Fe} + 6\text{HNO}_3 \text{ đặc} \longrightarrow \text{Fe}(\text{NO}_3)_3 + 3\text{NO}_2 + 3\text{H}_2\text{O}$.
753. $2\text{Fe} + 6\text{H}_2\text{SO}_4 \text{ đặc} \xrightarrow{t^\circ} \text{Fe}_2(\text{SO}_4)_3 + 3\text{SO}_2 + 6\text{H}_2\text{O}$.
754. $\text{Fe} + 4\text{HNO}_3 \text{ loãng} \longrightarrow \text{Fe}(\text{NO}_3)_3 + \text{NO} + 2\text{H}_2\text{O}$.
755. $3\text{Fe} + 2\text{O}_2 \xrightarrow{t^\circ} \text{Fe}_3\text{O}_4$.

B. HÓA HỮU CƠ.

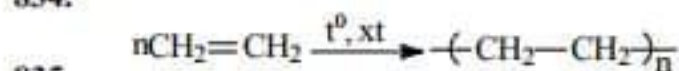
756. $2\text{C}_3\text{H}_8\text{O}_3 + \text{Cu}(\text{OH})_2 \longrightarrow (\text{C}_3\text{H}_7\text{O}_3)_2\text{Cu} + 2\text{H}_2\text{O}$
757. $2\text{C}_6\text{H}_{12}\text{O}_6 + \text{Cu}(\text{OH})_2 \longrightarrow (\text{C}_6\text{H}_{11}\text{O}_6)_2\text{Cu} + 2\text{H}_2\text{O}$

786. $C_nH_{2n+2} + mX_2 \xrightarrow{t, \text{ánh sáng}} C_nH_{2n+2-2m}X_m + mHX \uparrow$
787. $CH_3-CH_3 \xrightarrow{400^\circ C} CH_2=CH_2 + H_2 \uparrow$
788. $C_nH_{2n+2} \xrightarrow{\text{cracking}} C_mH_{2m} + C_xH_{2x+2}$
789. $C_nH_{2n+2} + (n+1)Cl_2 \xrightarrow{t, \text{ánh sáng}} nC + 2(n+1)HCl$
790. $C_nH_{2n+2} + (3n+1)/2 O_2 \xrightarrow{t, C} n CO_2 + (n+1)H_2O$
791. $C_2H_5-Cl + 2Na + Cl-CH_3 \longrightarrow C_2H_5-CH_3 + 2NaCl$
792. $R_1(COONa)_m + mNaOH_{(tr)} \xrightarrow{CaO, 400^\circ C} R_1H_m + mNa_2CO_3$
793. $CH_3COONa + NaOH \xrightarrow{CaO, 400^\circ C} CH_4 \uparrow + Na_2CO_3$
 $Al_4C_3 + 12 H_2O \longrightarrow 4Al(OH)_3 \downarrow + 3CH_4 \uparrow$
794. $CH_3-CH(CH_3)-CH_2-CH_3 + Cl_2 \xrightarrow{h\nu} CH_3-CCl-CH_2-CH_3$
795. $CH_2=CH_2 + Br_2 \rightarrow Br-CH_2-CH_2-Br$
796. $CH_2=CH_2 + HCl \rightarrow CH_3CH_2Cl$
797. $CH_2=CH_2 + H-OH \xrightarrow{t} HCH_2-CH_2OH$
798. $nCH_2=CH_2 \xrightarrow[100atm]{\text{pilot, } 100-300^\circ C} [-CH_2-CH_2-]_n$
799. $C_nH_{2n} + \frac{3n}{2} O_2 \xrightarrow{t} nCO_2 + nH_2O$
800. $3CH_2=CH_2 + 4H_2O + 2KMnO_4 \rightarrow 3HO-CH_2-CH_2-OH + 2MnO_2 \downarrow + 2KOH$
801. $CH_3CH_2OH \xrightarrow{H_2SO_4, 170^\circ C} CH_2=CH_2 + H_2O$
802. $CH_3-CH=CH_2 + H-Cl \rightarrow \begin{cases} CH_3-CHCl-CH_3 & (\text{sp chính}) \\ CH_3-CH_2-CH_2Cl & (\text{sp phụ}) \end{cases}$
803. $CH_2=CH-CH=CH_2 + 2H_2 \xrightarrow{Ni, t} CH_3-CH_2-CH_2-CH_3$
804. $CH_2=CH-CH=CH_2 + HBr (dd) \xrightarrow{40^\circ C} CH_3-CH=CH-CH_2Br$
805. $C_nH_{2n-2} + \frac{3n-1}{2} O_2 \rightarrow nCO_2 + (n-1)H_2O$
806. $CH_3-CH_2-CH_2-CH_3 \xrightarrow{400^\circ C} CH_2=CH-CH=CH_2 + 2H_2$
807. $CH \equiv CH + HCl \xrightarrow[150-200^\circ C]{HgCl_2} CH_2=CH-Cl : \text{vinyl clorua}$
808. $CH \equiv CH + H-OH \xrightarrow{HgSO_4} CH_2=CH-OH \rightarrow CH_3CHO$
809. $2 CH \equiv CH \xrightarrow{400^\circ C} CH \equiv C-CH \equiv CH_2$

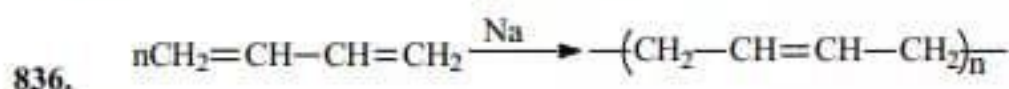




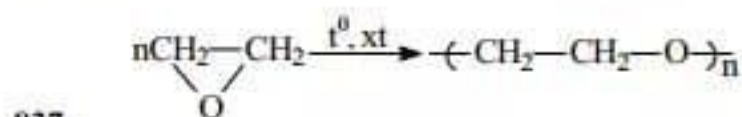
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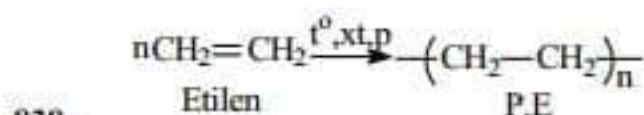
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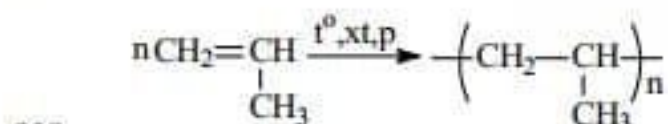
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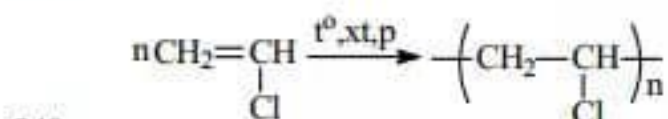
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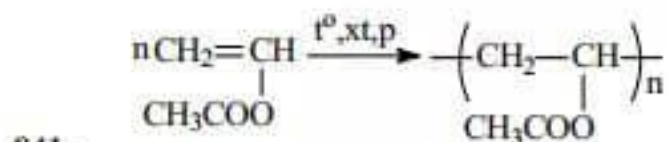
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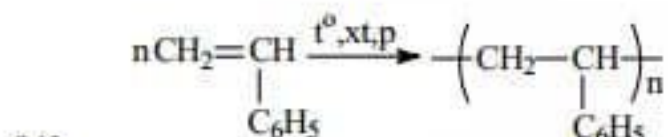
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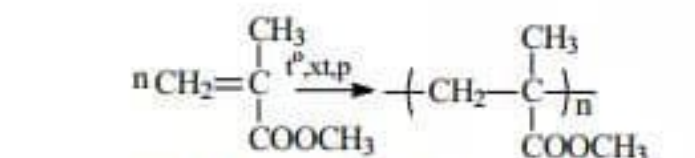
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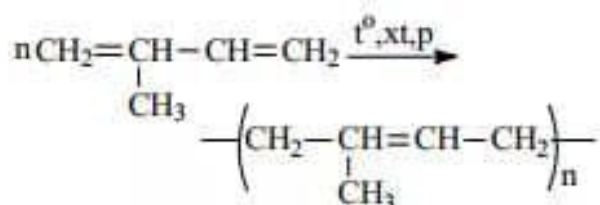
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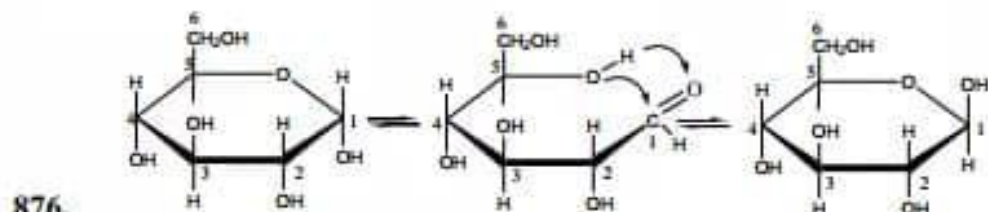
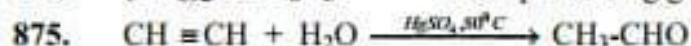
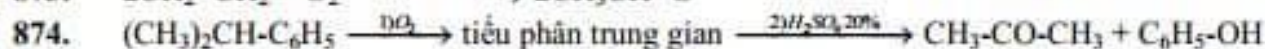
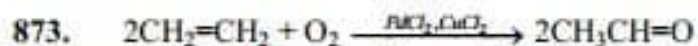
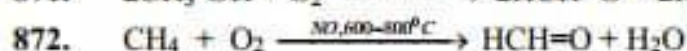
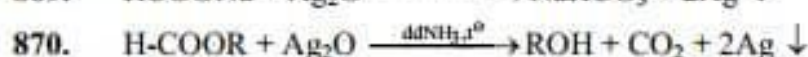
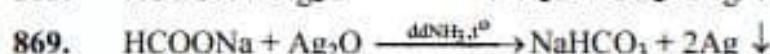
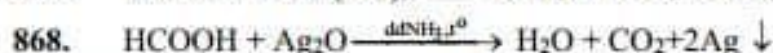
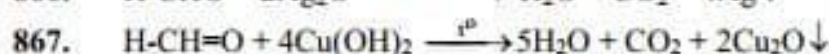
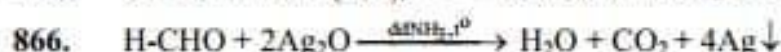
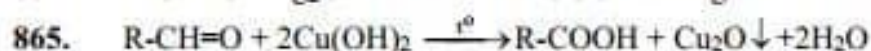
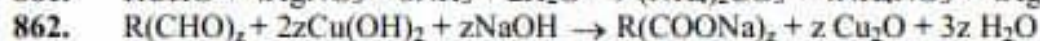
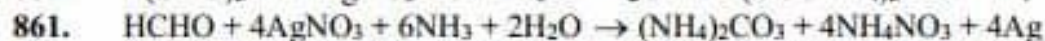
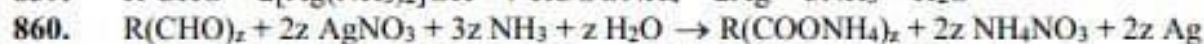
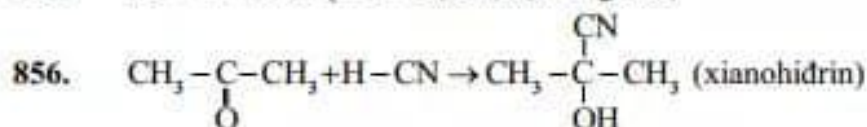
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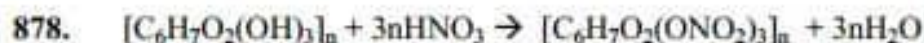
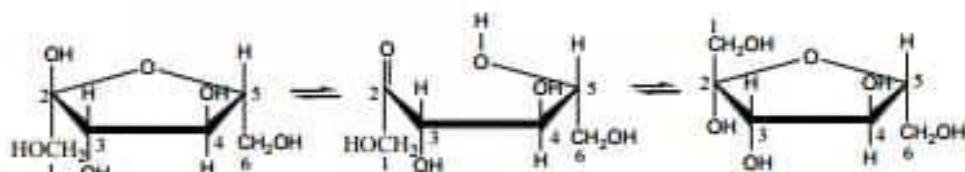
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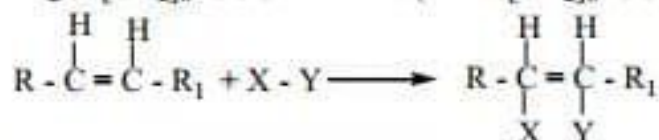
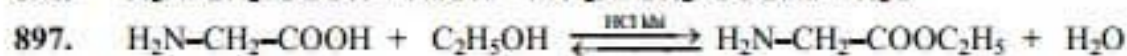
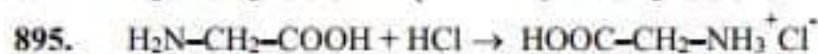
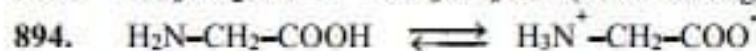
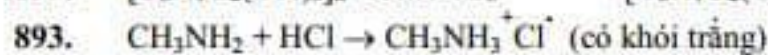
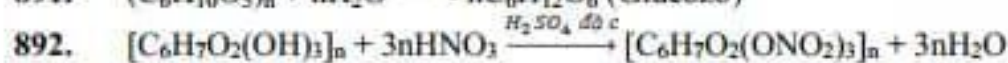
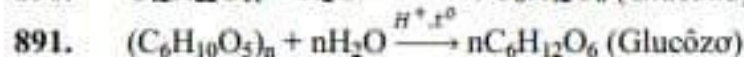
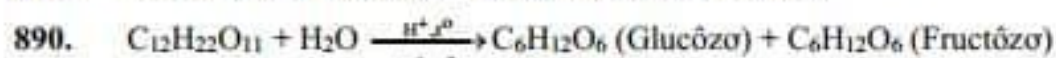
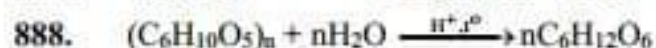
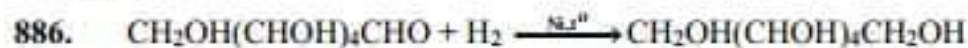
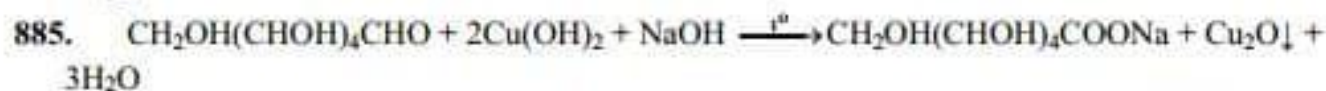
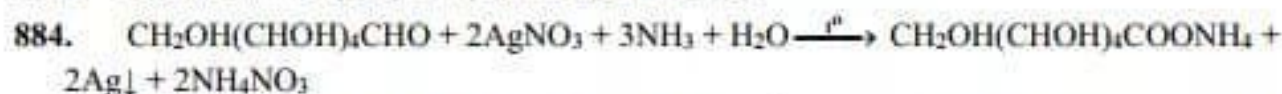
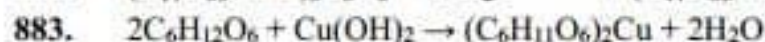
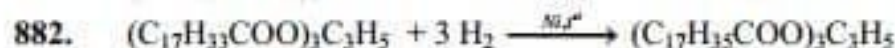
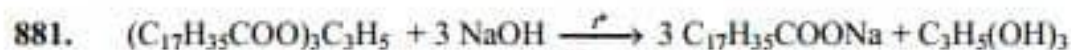
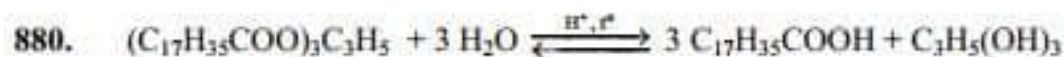


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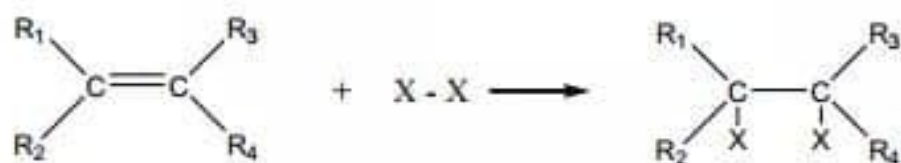


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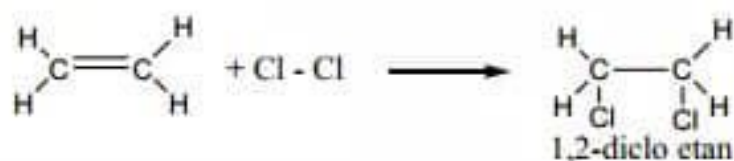




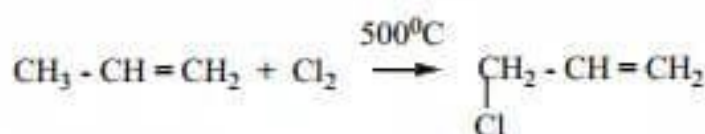
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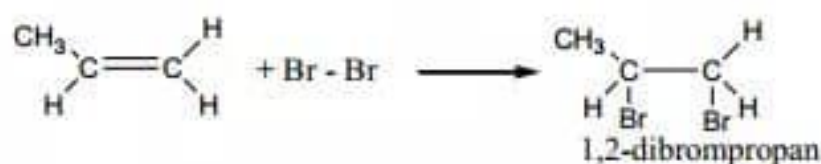
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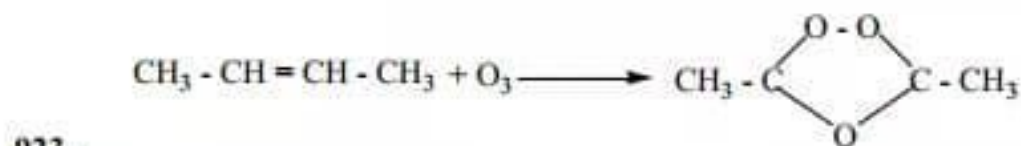
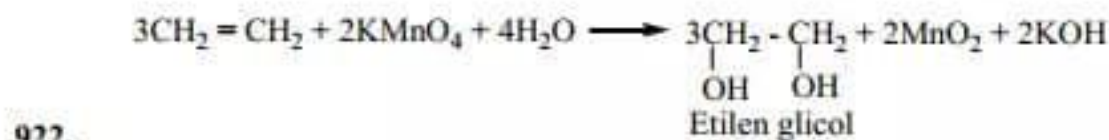
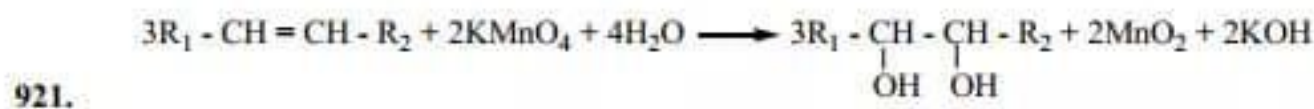
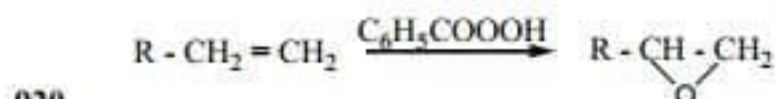
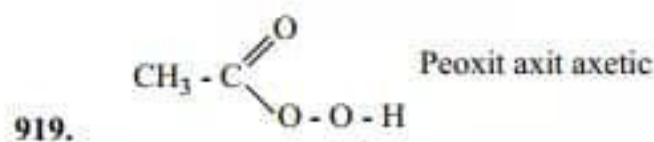
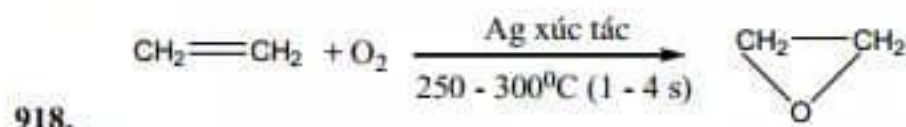
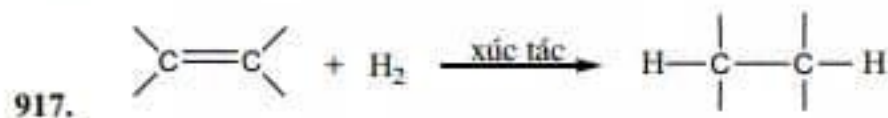
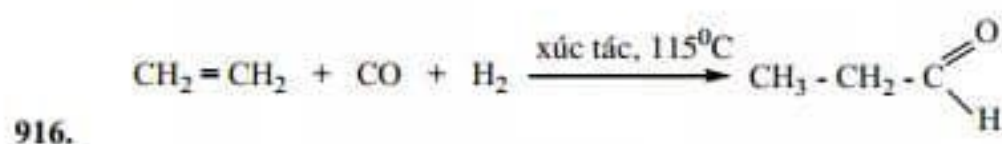
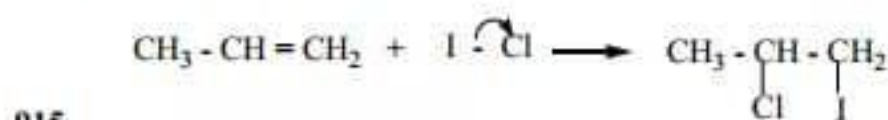
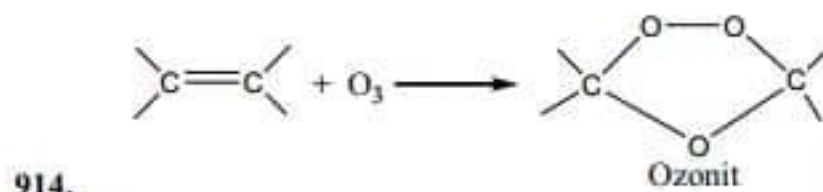
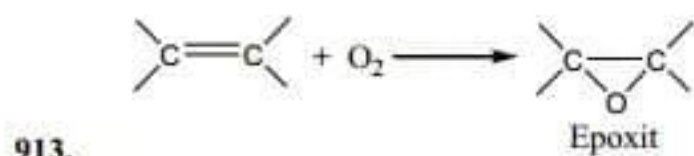
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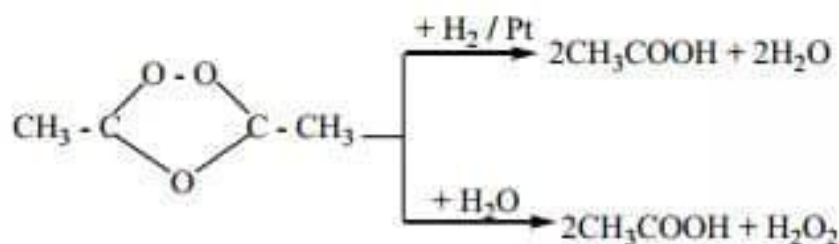


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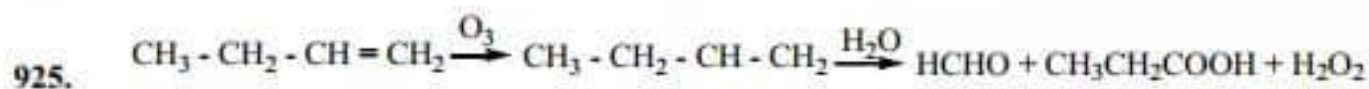


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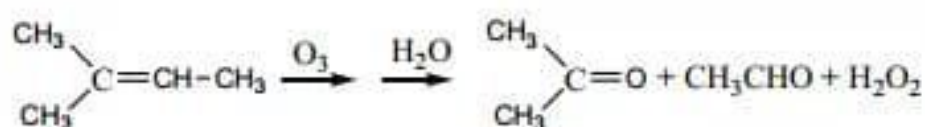




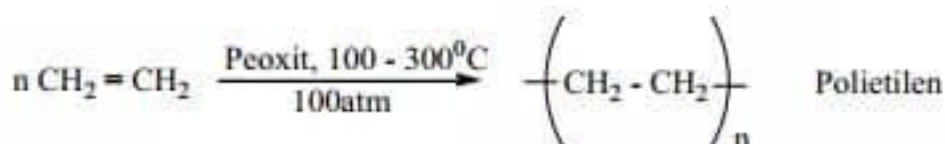
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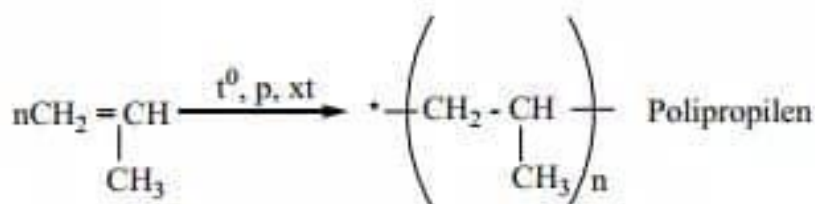
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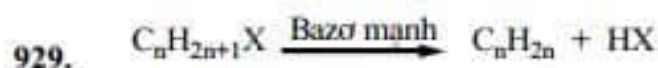
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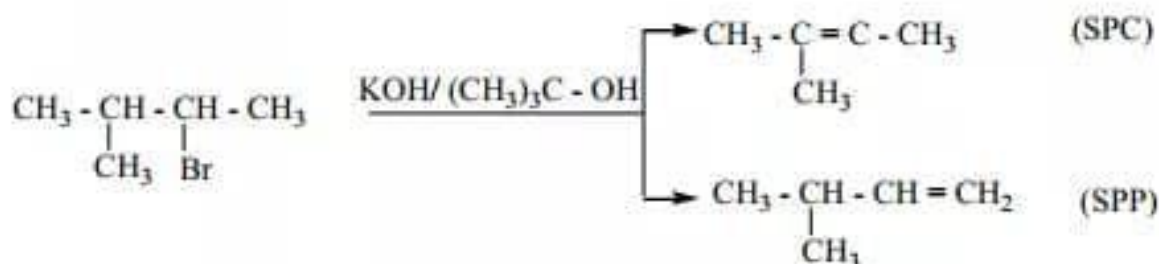
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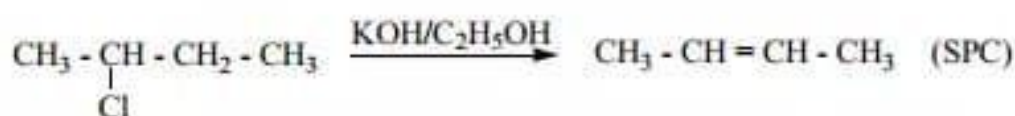
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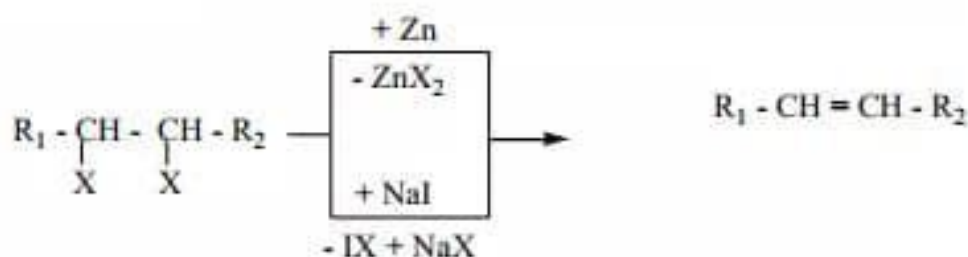
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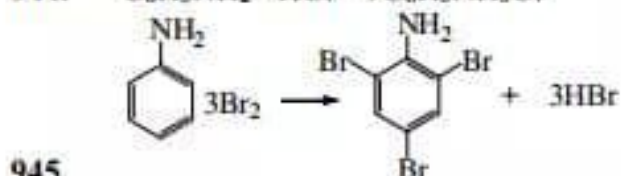
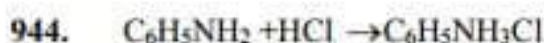
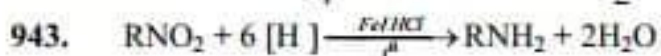
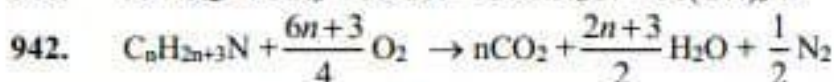
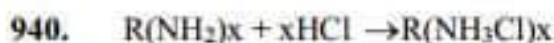
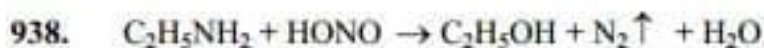
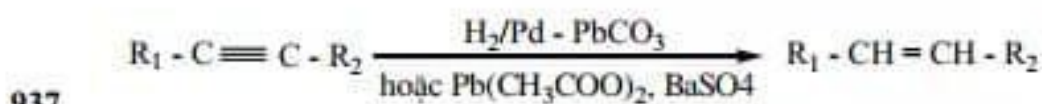
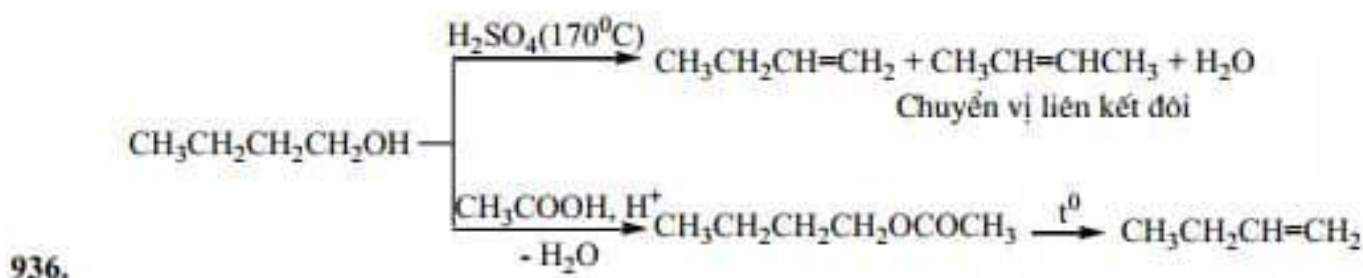
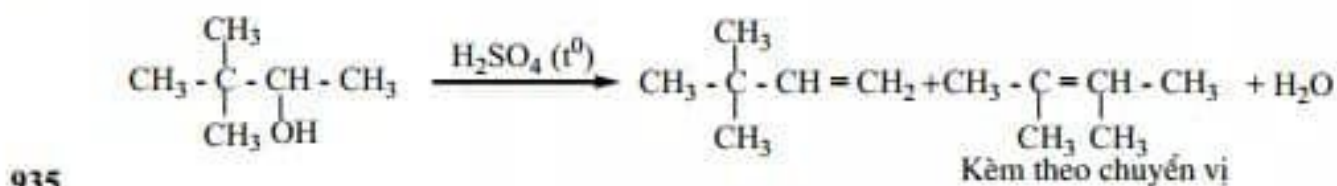
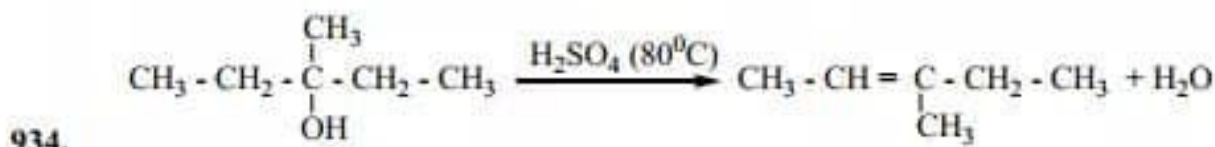
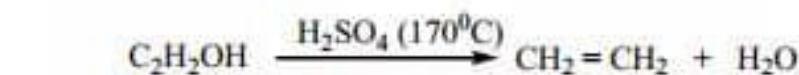
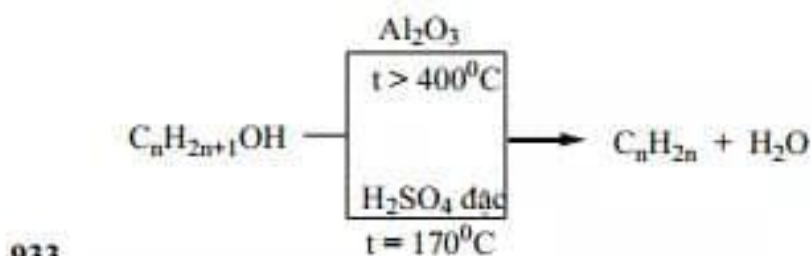
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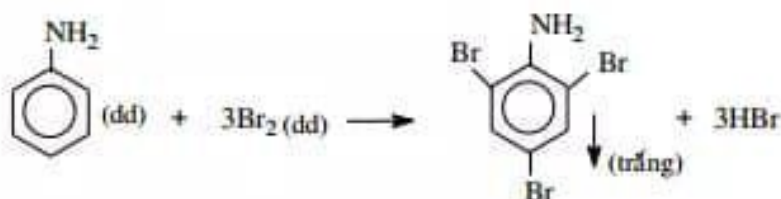
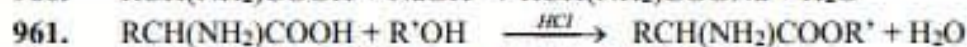
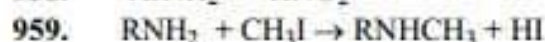
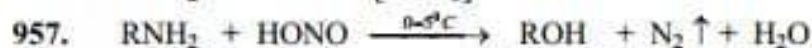
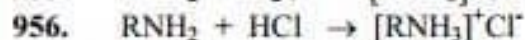
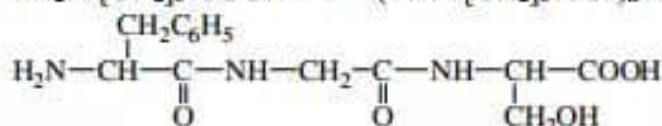
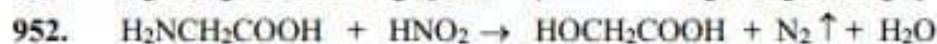
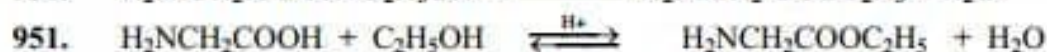
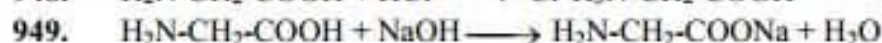
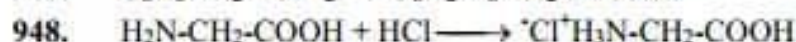


931.



932.





964.

